



KOZEN Financial
SDK Development Documentation
V1.3

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● 1. Revision History

Version	Release	Modify Record	Adapted SDK version	Author
1.3	2025/08/22	✓ Add explanations for three DEKPT_DES functions ✓ Include instructions for MK/SK writing ✓ Fix the issue with ConstantSecurity in the previous version of the document.	FinnancialService1.1.x	Johnny
1.2	2025/07/24	✓ Add NFC TAG, ECR (only includes serial port & USB-to-serial), Felica card support, front and rear camera scanning ✓ Support for separate updates of the Kozen SDK ✓ Fix the API level of the system environment to 23. ✓ getCardExistStatus adds a return value for card presence.	FinnancialService1.1.x	Johnny
1.1	2025/03/31	✓ Supplemental printing & barcode module error codes/constants ✓ Remove duplicate error messages in EMV module - Add Pinpad rotation support	FinnancialService1.0.x	Yue.Cui Yao.Zhang Tong.Liu Johnny
1.0	2025/03/10	✓ Add error code definition, entity definition, access permission ✓ Add EMV module ✓ Add SDK integration description	FinnancialService1.0.x	Yao.zhang Sunan Johnny
0.5	2025/03/03	✓ Add a description of the financial SDK engine module	FinnancialService1.0.x	Sunan Johnny
0.4	2025/02/27	✓ Update card reader, password keyboard, and some interfaces for general operations	FinnancialService1.0.x	Johnny
0.3	2025/1/24	✓ Add EMV module, printer module, scanner module, and security module ✓ Update the API description for card reader, PINPAD, and general operation ✓ Add card detection and card positioning functions; Add PINPAD parameter configuration	FinnancialService1.0.x	Johnny
0.2	2025/1/2	✓ Add object description	FinnancialService1.0.x	Johnny
0.1	2024/11/12	✓ Initial version	FinnancialService1.0.x	Johnny

● 2. Overview

2.1 Introduction

KozenFinancialService is a hardware firmware-based API SDK provided by KOZEN. Designed specifically for Java and Android developers. This SDK enables developers to quickly access hardware operation interfaces for KOZEN financial terminal, facilitating efficient business logic implementation.

The SDK primarily includes the following modules: Basic Device Information, Card Operations, PIN Pad, EMV, and Security mode.

This document serves as the KozenFinancialService API Reference.

2.2 Android version and IDE version supported by the SDK

System environment	Platform	Compile environment
Android 6.0 and above	ARM 64, ARM 32	Android Studio, IntelliJ

2.3 Feature Introduction

2.3.1 Financial SDK Engine Module

- This module handles SDK initialization and provides access to various module operation classes.
- Operation class object: FinancialEngine

2.3.2 Cardreader Module

- This module handles card reader functionality.
- Operation class object: ICardReaderManager
- Example to get the module operation class:
JAVA: FinancialEngine.INSTANCE.getCardReaderManager()
Kotlin: FinancialEngine.cardReaderManager()

2.3.3 EMV Module

- This module handles EMV functionality.
- Operation class object: IEmvManager
- Example to get the module operation class:
JAVA: FinancialEngine.INSTANCE.getEmvManager()
Kotlin: FinancialEngine.emvManager()

2.3.4. General Module

- This module handles basic device control functionalities.
- Operation class object: IGeneralManager

- Example to get the module operation class:

JAVA: `FinancialEngine.INSTANCE.getGeneralManager()`

Kotlin: `FinancialEngine.generalManager()`

2.3.5. Pinpad Module

- This module handles Pinpad functionality.
- Operation class object: `IPinpadManager`
- Example to get the module operation class:

JAVA: `FinancialEngine.INSTANCE.getPinpadManager()`

Kotlin: `FinancialEngine.pinpadManager()`

2.3.6. Printer Module

- This module handles printing functionality.
- Operation class object: `IPrinterManager`
- Example to get the module operation class:

JAVA: `FinancialEngine.INSTANCE.getPrinterManager()`

Kotlin: `FinancialEngine.printerManager()`

2.3.7. Scanner Module

- This module handles scanning functionality.
- Operation class object: `IScannerManager`
- Example to get the module operation class:

JAVA: `FinancialEngine.INSTANCE.getScannerManager()`

Kotlin: `FinancialEngine.scannerManager()`

2.3.8. Security Module

- This module handles encryption/decryption algorithms and key-related functionalities.
- Operation class object: `ISecurityManager`
- Example to get the module operation class:

JAVA: `FinancialEngine.INSTANCE.getSecurityManager()`

Kotlin: `FinancialEngine.securityManager()`

2.4 Importing the Financial SDK

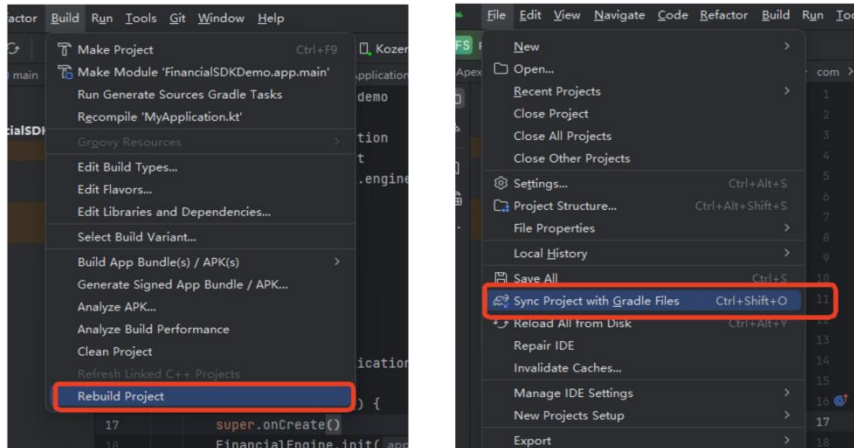
Local Dependency: Place the `FinancialLib-x.x.x-release.aar` file in the `libs` directory of your Android Studio project. Add the following code to the `build.gradle` file:

kotlin

复制

```
dependencies {
    implementation(files("libs/FinancialLib-1.0.0-release.aar"))
}
```

After importing the `.aar` file, sync and rebuild the project.



2.5 Initializing the Financial SDK

Please initialize the Financial SDK in your Application. Example:

kotlin

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```
class DemoApplication : Application() {

    override fun onCreate() {
        super.onCreate()

        /**
         * Example: Financial service initialization
         */
        FinancialEngine.init(this, object : InitListener {
            override fun onResult(result: Int, errorMsg: String?) {
                Toast.makeText(
                    this@DemoApplication,
                    if (result == 0) "success" else "failed, $errorMsg",
                    Toast.LENGTH_SHORT
                ).show()
            }
        })
    }
}
```

If initialization is successful, the callback will return `result == 0`.

If initialization fails, the callback will return `result == -1`.

After the Financial SDK is successfully initialized, use `FinancialEngine` to obtain the operation objects for each module. Otherwise, an error code `-10001` (Financial service not connected) will be thrown.

If the financial service is disconnected during use, the InitListener will be called again with result == -1. Upon receiving this callback, reinitialize the Financial SDK.

If an interface throws the error code -10001 (Financial service not connected) during use, reinitialize the Financial SDK.

● 3. API Interface Introduction

void init(android.content.Context application, InitListener callback)	Financial SDK initialization
ICardReaderManager getCardReaderManager()	Card Reader Operation module
IEmvManager getEmvManager()	EMV Operation module
IGeneralManager getGeneralManager()	Device Basic Operation function
IPinpadManager getPinpadManager()	PINPAD Operation module
IPrinterManager getPrinterManager()	Printing Operation module
ISecurityManager getSecurityManager()	Security module
IEcrManager getEcrManager()	ECR Operation module
IScannerManager getScannerManager()	Scanner Operation module

➤ 3.1 Financial SDK initialization

3.1.1 Initialize the FinancialService instance

Prototype	void init(android.content.Context application, InitListener callback)
Function	Initialize the FinancialService instance
Parameters	Parameters: application – context callback – initialization callback
Return value	
Notes	

-- FinancialService instance callback – InitListener --

void onResult(int result, String errorMsg)	Initialization results
--	------------------------

3.1.2 Initialization results

Prototype	void onResult(int result, String errorMsg)
Function	Initialization results
Parameters	Parameters: result – initialization result 0: success Others: failure – more details to see CommonError errorMsg – error message
Return value	
Notes	

➤ 3.2 Scanner Operation module

-- Get scanner operation module - getScannerManager --

void close()	Close the scanner module.
boolean isBarcodeEnabled(ConstantScanner.BarcodeFormat type)	Check whether a certain barcode type can be recognized.
void open(IConnectionStatusListener callback)	Open the scanner module.
int registerResultCallback(IScannerResultCallback callback)	Register the callback listener for the barcode scanning result.
int setBarcodeEnable(boolean enable)	Enable/Disable all supported barcode types.
int setBarcodeEnable(List<ConstantScanner.BarcodeFormat> types, boolean enable)	Enable/Disable specified barcode type to be recognized.
int startScan()	Trigger the barcode scanning action and start scanning.
int stopScan()	Stop scanning.

3.2.1 Open the scanner module.

Prototype	void open(IConnectionStatusListener callback)
Function	Open the barcode scanning module Initialize the barcode scanning module. Only after successful called, other interfaces can be available.
Parameters	Parameters: callback - connection status callback
Return value	
Notes	

3.2.2 Close the scanner module

Prototype	void close()
Function	Close the barcode scanning module. When the barcode scanning module is no longer needed, use this interface to release the resources
Parameters	
Return value	
Notes	

3.2.3 Register the callback listener for the code scanning result

Prototype	int registerResultCallback(IScannerResultCallback callback)
Function	Register the callback listener for the scan result. After successful scanning, the barcode type and scan result will be returned through this callback. It needs to be called after OPEN API.
Parameters	Parameters: callback - callback for scanning result
Return value	Return: 0: The operation is successfully executed;

	Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in ScannerError and CommonError
Notes	

3.2.4 Trigger the barcode scanning action and start scanning

Prototype	int startScan()
Function	Trigger the barcode scanning action and start scanning. It needs to be called after OPEN API.
Parameters	
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in ScannerError and CommonError
Notes	

3.2.5 Stop scanning

Prototype	int stopScan()
Function	Stop scanning. It needs to be called after OPEN API.
Parameters	
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in ScannerError and CommonError
Notes	

3.2.6 Enable/Disable specified barcode type to be recognized

Prototype	int setBarcodeEnable(List<ConstantScanner.BarcodeFormat> types, boolean enable)
Function	Enable/disable support for the specified type of barcode. When support is enabled, the scanner can recognize the barcode. Otherwise, it is disabled. This function needs to be called after the OPEN API is called.
Parameters	Parameters: types - An array of ConstantScanner.BarcodeFormat enumeration types, used to specify the barcode types to be enabled or disabled. enable - true: enable support; false: disable support.
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in ScannerError and CommonError
Notes	

3.2.7 Enable/Disable all supported barcode types.

Prototype	int setBarcodeEnable(boolean enable)
Function	Enable/Disable all supported barcode types. This function needs to be called after the OPEN API is called.
Parameters	Parameters: enable – true: Enable all code system support; false: Disable all code system support. The scan button will light up at this time, but no barcode can be recognized
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in ScannerError and CommonError
Notes	

3.2.8 Check whether a certain barcode type can be recognized.

Prototype	boolean isBarcodeEnabled(ConstantScanner.BarcodeFormat type)
Function	Check whether a certain barcode type can be recognized. This function needs to be called after the OPEN API is called.
Parameters	Parameters: type – code type
Return value	Return: true: The scanner can recognize the barcode type; false: The scanner cannot recognize the barcode type. Please note that false will also be returned if the SDK status is abnormal.
Notes	

-- Scanner connection status listener – IConnectionStatusListener --

void onConnected()	Monitoring barcode scanning service is connected
void onDisconnected()	Monitoring barcode scanning service is disconnected
void onError(int error, String msg)	Monitoring barcode scanning service error messages

3.2.9 Monitoring barcode scanning service is connected

Prototype	void onConnected()
Function	Monitoring barcode scanning service is connected
Parameters	
Return value	
Notes	Will be called when a connection is established with the code scanning service, indicating that the code scanning module has been initialized successfully

3.2.10 Monitoring barcode scanning service is disconnected

Prototype	void onDisconnected()
-----------	-----------------------

Function	Monitoring barcode scanning service is disconnected
Parameters	
Return value	
Notes	Will be called when the connection with the code scanning service is lost

3.2.11 Monitoring barcode scanning service error messages

Prototype	void onError(int error, String msg)
Function	Monitoring barcode scanning service error messages
Parameters	Parameters: error – error code. For the specific meaning of the error code, please refer to the definitions in the ScannerError and CommonError msg – error description
Return value	
Notes	

-- Scanner Result Callback – IScannerResultCallback --

void onResult(String sym, String barcode)	Get the scan result and code type
---	-----------------------------------

3.2.12 Get the scan result and code type

Prototype	void onResult(String sym, String barcode)
Function	Get the scan result and code type
Parameters	Parameters: sym – code type barcode – scan result
Return value	
Notes	

➤ 3.3 Card Reader Operation module

-- Get card reader operation module – getCardReaderManager --

int powerOff(int cardType)	Power off the card
int powerOn(int cardType)	Power on the card
int checkCard(int cardType, int timeout, ICheckCardListener callback)	Start checking card
int getCardExistStatus(int cardType)	Check whether the card is in the card reader slot
int stopCheck()	Stop checking card
Int transmitApdu(int cardType, byte[] sendBuf, CustomByteArray rspBuf,	Transmit APDU command to card

CustomByteArray swBuf)	
int hceWrite(android.nfc.NdefMessage msg, int timeout)	Write NDEF data via HCE

3.3.1 Power on the card

Prototype	int powerOn(int cardType)
Function	Card reader powered on
Parameters	Parameters: cardType – card type 1. Supports single card and multiple card types. If input ConstantCardReader.CardType.ALL, can detect all card types. 2. Multiple card types are detected such as follows: ConstantCardReader.CardType.CONTACT ConstantCardReader.CardType.MAGNETIC 3. More card type details in ConstantCardReader.CardType
Return value	Returns: 0: Success Non-0: Failure – see CardReaderError
Notes	Note: 1. Due to protocol conflicts, Felica cards cannot be powered on simultaneously with other contactless cards. 2. When powering on cards, if the parameter ConstantCardReader.CardType.ALL is passed, only other contactless cards will be powered on by default, excluding Felica cards. 3. When powering on other contactless cards, Felica cards will be powered off; likewise, when powering on Felica cards, other contactless cards will be powered off.

3.3.2 Power off the card

Prototype	int powerOff(int cardType)
Function	Power off the card
Parameters	Parameters: cardType – card type 1. Supports single card and multiple card types. If input ConstantCardReader.CardType.ALL, can detect all card types. 2. Multiple card types are detected such as follows: ConstantCardReader.CardType.CONTACT ConstantCardReader.CardType.MAGNETIC 3. More card type details in ConstantCardReader.CardType
Return value	Returns: 0: Success Non-0: Failure – see CardReaderError
Notes	

3.3.3 Start checking card

Prototype	int checkCard(int cardType, int timeout,
-----------	---

	ICheckCardListener callback)
Function	Start checking card
Parameters	Parameters: cardType – card type 1. Supports single card and multiple card types. If input ConstantCardReader.CardType.ALL, can detect all card types. 2. Multiple card types are detected such as follows: ConstantCardReader.CardType.CONTACT ConstantCardReader.CardType.MAGNETIC 3. If both CardType.ALL and specific card types are passed, the specific card types take precedence. Example: ConstantCardReader.CardType.ALL ConstantCardReader.CardType.CONTACT will only detect contact cards during card detection. 4. More card type details in ConstantCardReader.CardType timeout – timeout, in seconds callback – card check callback
Return value	Returns: 0: Success Non-0: Failure – see CardReaderError
Notes	Note: 1. The card will not be powered off automatically after the card checking is completed. Please note that the card must be powered off manually after the checking is completed; 2. During the card checking process, a card checking error will not terminate the process, but the onError callback will be called; 3. The checking process will continue until one type of card is detected or checking time out. 4. There is a protocol conflict between contactless cards and Felica cards; they cannot be detected simultaneously. 5. If the card type is set to detect all cards (ConstantCardReader.CardType.ALL), felica cards will not be detected by default.

3.3.4 Stop checking card

Prototype	int stopCheck()
Function	Stop checking card
Parameters	
Return value	Returns: 0: Success Non-0: Failure – see CardReaderError
Notes	Note: 1. If you want to interrupt checking manually during the card checking process, please call this method to end the card inspection 2. In the normal process, such as card is found or timeout occurs, the process will automatically stop 3. Stop the card checking process will not power off card; 4. After this function, please use the poweroff function to avoid affecting the power consumption of the machine.

3.3.5 Check whether the card is in the card reader slot

Prototype	int getCardExistStatus(int cardType)
Function	Check whether the card is in the card reader slot
Parameters	cardType – card type 1. Only supports single card 2. Supported card types: 2.1 ConstantCardReader.CardType.CONTACT – contact card 2.2 ConstantCardReader.CardType.CONTACTLESS – contactless card 2.3 ConstantCardReader.CardType.FELICA – felica card
Return value	Return: 0: Card present -1: Card not present Other: Failure – see CardReaderError for details
Notes	Note: 1. This function does not support multiple cards, only single card types; 2. This function will power on the checked card, and will not automatically power off after the detection; 3. After this function, please use the poweroff function to avoid affecting the power consumption of the machine.

3.3.6 Transmit APDU command to card

Prototype	int transmitApdu(int cardType, byte[] sendBuff, CustomByteArray rspBuf, CustomByteArray swBuf)
Function	Transmit APDU command to card
Parameters	Parameters: cardType – card type 1. Only supports single card 2. Supported card types: 2.1 ConstantCardReader.CardType.CONTACT – contact card 2.2 ConstantCardReader.CardType.CONTACTLESS – contactless card 2.3 ConstantCardReader.CardType.FELICA – felica card sendBuff – data to be transparently transmitted to the card, maximum 1929B rspBuf – card response data swBuf – card response data – software version number
Return value	Return: 0: Card is in the slot Others: Card is not in slot, see CardReaderError
Notes	Note: 1. This function does not support multiple cards, only single card types; 2. This function will power on the checked card automatically , but will not power off card after the detection is completed;

	3. After this function, please use the poweroff function to avoid affecting the power consumption of the machine. 4. APDU command transmission is allowed only after successful card check; this method will fail if the card is powered on without card check
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3.3.7 write HCE data via NFC tag

Prototype	int hceWrite(android.nfc.NdefMessage msg, int timeout)
Function	Write NDEF data via HCE
Parameters	Parameters: msg – Data of type NdefMessage , max 255 bytes timeout – Timeout in seconds (timeout <= 0 means no timeout)
Return Value	Return: 0 – Success; Others – Failure. Refer to CardReaderError for details
Notes	This method writes NDEF data using Host Card Emulation.

-- Card check listener – ICheckCardListener --

void findContactCard(android.os.Bundle info)	Contact card found successfully
void findMagstripeCard(android.os.Bundle info)	Magstripe card found successfully
void findContactlessCard(android.os.Bundle info)	Contactless card found successfully
void findFelicaCard(android.os.Bundle info)	Felica card found successfully
void onError(int code, String message)	Card detection error
void onTimeout()	Card detection timeout

3.3.7 Magstripe card found successfully

Prototype	void findMagstripeCard(android.os.Bundle info)
Function	Magstripe card found successfully
Parameters	Parameters: info – return data
Return value	
Notes	Parameter constant value The following data will be returned during the card detection process ConstantCardReader.CARD_CHANNEL – logical channel number ConstantCardReader.CARD_SERIAL_NUM – Card serial number ConstantCardReader.CARD_ATTRIBUTE – ATR ConstantCardReader.TRACK1 – Track 1 data ConstantCardReader.TRACK2 – Track 2 data ConstantCardReader.TRACK3 – Track 3 data

3.3.8 Contact card found successfully

Prototype	void findContactCard(android.os.Bundle info)
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Function	Contact card found successfully
Parameters	Parameters: info – return data
Return value	
Notes	Parameter constant value The following data will be returned during the card detection process ConstantCardReader.CARD_CHANNEL – logical channel number ConstantCardReader.CARD_SERIAL_NUM – Card serial number ConstantCardReader.CARD_ATTRIBUTE – ATR

3.3.9 Contactless card found successfully

Prototype	void findContactlessCard(android.os.Bundle info)
Function	Contactless card found successfully
Parameters	Parameters: info – return data
Return value	
Notes	Parameter constant value The following data will be returned during the card detection process ConstantCardReader.CARD_CHANNEL – logical channel number ConstantCardReader.CARD_SERIAL_NUM – Card serial number ConstantCardReader.CARD_ATTRIBUTE – ATR

3.3.10 Felica card found successfully

Prototype	void findFelicaCard(android.os.Bundle info)
Function	Felica card found successfully
Parameters	Parameters: info – return data
Return Value	
Notes	Parameter constant value The following data will be returned during the card detection process ConstantCardReader.ID_FOR_MANUFACTURER – Unique card ConstantCardReader.PARAMETER_FOR_MANUFACTURER – Card parameters ConstantCardReader.REQUEST_DATA – Command response

3.3.11 Card detection error

Prototype	void onError(int code, String message)
Function	Card detection error
Parameters	Parameters: code – error code, more details to see CardReaderError message – error description
Return value	

Notes	
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3.3.12 Card detection timeout

Prototype	void onTimeout()
Function	Card detection timeout
Parameters	
Return value	
Notes	

➤ 3.4 EMV Operation module

-- Get EMV operation module - getEmvManager --

int deleteAid()	Delete AID
int deleteAppleMerchant()	Delete Apple Merchant transaction parameters
int deleteCapk()	Delete CAPK
int deleteDRL(int type)	Delete Dynamic Reader Limits (DRL) configuration parameters
int deleteExceptionFile()	Delete Exception File
int deleteRevocationIPK()	Delete Revocation IPK
int deleteService()	Delete RuPay payment terminal parameters
List<EmvAid> getAid()	Get AID list
int getAppleMerchant(android.os.Bundle bundle)	Get Apple Merchant transaction parameters
int getAppleTerminal(android.os.Bundle bundle)	Get Apple VAS transaction parameters
List<EmvCapk> getCapk()	Get CAPK list
int getDRL(int type, android.os.Bundle bundle)	Get Dynamic Reader Limits (DRL) configuration parameters
List<EmvExceptionFile> getExceptionFile()	Get configured Exception File
byte[] getKernel(String[] tags)	Get EMV Tag/Object parameters
List<EmvRevocationIPK> getRevocationIPK()	Get configured CAPK Revocation parameters
int getService(android.os.Bundle bundle)	Get RuPay payment terminal parameters
int getTerminal(int type, android.os.Bundle bundle)	Get EMV kernel terminal information
String getVersion(int type)	Get Kernel version information
int setAid(EmvAid emvAid)	Load AID
int setAppleMerchant(android.os.Bundle bundle)	Set Apple Merchant transaction parameters
int setAppleTerminal(android.os.Bundle bundle)	Set Apple VAS transaction parameters
int setCapk(EmvCapk emvCapk)	Add CAPK
int setCardInfoResponse(android.os.Bundle bundle)	Set card information confirmation result
int setDRL(int type, android.os.Bundle bundle)	Set Dynamic Reader Limits (DRL) configuration parameters
int setExceptionFile(EmvExceptionFile exceptionFile)	Set Exception File
int setKernel(byte[] tlv)	Set EMV Tag/Object parameters
int setOnlineResponse(android.os.Bundle bundle)	Set online transaction result

int setPinResponse(android.os.Bundle bundle)	Set PIN input result
int setRevocationIPK(EmvRevocationIPK revocationIPK)	Set Revocation IPK
int setSelectApplicationResponse(int position)	Set multiple application selection result
int setService(android.os.Bundle bundle)	Set RuPay terminal parameters
int setTerminal(int type, android.os.Bundle bundle)	Set EMV kernel terminal information
int startTransaction(android.os.Bundle bundle, IEmvListener callback)	Start EMV transaction
int stopTransaction()	Stop transaction

3.4.1 Set the terminal information for the EMV kernel

Prototype	int setTerminal(int type, android.os.Bundle bundle)
Function	Set the terminal information for the EMV kernel.
Parameters	type: Terminal type, with the following enumerated values: ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_TERMINAL ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_CONFIG ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_VISA ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_UNIONPAY ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_MASTERCARD ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_DISCOVER ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_AMEX ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_MIR ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_RUPAY ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_INTERAC bundle: Terminal parameters. For constant details, refer to ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.2 Get the terminal information for the EMV kernel

Prototype	int getTerminal(int type, android.os.Bundle bundle)
Function	Get the terminal information for the EMV kernel.
Parameters	type: Terminal type, with the following enumerated values: ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_TERMINAL ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_CONFIG ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_VISA ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_UNIONPAY ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_MASTERCARD ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_DISCOVER

	ConstantEnv.POIEnvCoreManager.EnvTerminalConstraints.TYPE_AMEX ConstantEnv.POIEnvCoreManager.EnvTerminalConstraints.TYPE_MIR ConstantEnv.POIEnvCoreManager.EnvTerminalConstraints.TYPE_RUPAY ConstantEnv.POIEnvCoreManager.EnvTerminalConstraints.TYPE_INTERAC bundle: Terminal parameters. For details, refer to the class definition of ConstantEnv.POIEnvCoreManager.EnvTerminalConstraints.
Return value	0: Success Others: Failure. Refer to EnvError for error codes.
Notes	

3.4.3 Set AID

Prototype	int setAid(EnvAid envAid)
Function	Set AID
Parameters	envAid: Parameter entity. For details, refer to EnvAid.
Return value	0: Success Others: Failure. Refer to EnvError for error codes.
Notes	

3.4.4 Delete AID

Prototype	int deleteAid()
Function	Delete AID
Parameters	None
Return value	0: Success Others: Failure. Refer to EnvError for error codes.
Notes	

3.4.5 Retrieve AID List

Prototype	List<EnvAid> getAid()
Function	Retrieve the AID list
Parameters	None
Return value	List of EnvAid. For details, refer to EnvAid.
Notes	

3.4.6 Add CAPK

Prototype	int setCapk(EnvCapk envCapk)
Function	Add CAPK
Parameters	envCapk: Parameter entity. For details, refer to EnvCapk.
Return value	0: Success Others: Failure. Refer to EnvError for error codes.
Notes	

3.4.7 Delete CAPK

Prototype	int deleteCapk()
Function	Delete CAPK
Parameters	None
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.8 Retrieve the CAPK list

Prototype	List<EmvCapk> getCapk()
Function	Retrieve the CAPK list
Parameters	None
Return value	List of EmvCapk. For details, refer to EmvCapk.
Notes	

3.4.9 Set Exception File

Prototype	int setExceptionFile(EmvExceptionFile exceptionFile)
Function	Set Exception File
Parameters	exceptionFile: Parameter entity. For details, refer to EmvExceptionFile.
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.10 Delete Exception File

Prototype	int deleteExceptionFile()
Function	Delete Exception File
Parameters	None
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.11 Retrieve the configured Exception File

Prototype	List<EmvExceptionFile> getExceptionFile()
Function	Retrieve the configured Exception File
Parameters	None
Return value	List of EmvExceptionFile. For details, refer to EmvExceptionFile.
Notes	

3.4.12 Set RevocationIPK

Prototype	int setRevocationIPK(EmvRevocationIPK revocationIPK)
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Function	Set RevocationIPK
Parameters	revocationIPK: Parameter entity. For details, refer to EmvRevocationIPK.
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.13 Delete RevocationIPK

Prototype	int deleteRevocationIPK()
Function	Delete RevocationIPK
Parameters	None
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.14 Retrieve the configured CAPK Revocation parameters

Prototype	List<EmvRevocationIPK> getRevocationIPK()
Function	Retrieve the configured CAPK Revocation parameters
Parameters	None
Return value	List of EmvRevocationIPK. For details, refer to EmvRevocationIPK.
Notes	

3.4.15 Set Dynamic Reader Limits (DRL) configuration parameters

Prototype	int setDRL(int type, android.os.Bundle bundle)
Function	Set Dynamic Reader Limits (DRL) configuration parameters
Parameters	type: DRL type. Supported card scheme types are as follows: ConstantEmv.POIEmvCoreManager.EmvDrlConstraints.TYPE_VISA ConstantEmv.POIEmvCoreManager.EmvDrlConstraints.TYPE_AMEX bundle: Parameter values. For details, refer to ConstantEmv.POIEmvCoreManager.EmvDrlConstraints.
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.16 Delete Dynamic Reader Limits (DRL) configuration parameters

Prototype	int deleteDRL(int type)
Function	Delete Dynamic Reader Limits (DRL) configuration parameters
Parameters	type: DRL type. Supported card scheme types are as follows: ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_VISA ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_AMEX
Return value	0: Success

	Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.17 Retrieve Dynamic Reader Limits (DRL) configuration parameters

Prototype	int getDRL(int type, android.os.Bundle bundle)
Function	Retrieve Dynamic Reader Limits (DRL) configuration parameters
Parameters	type: DRL type. Supported card scheme types are as follows: ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_VISA ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints.TYPE_AMEX bundle: Parameter values. For details, refer to ConstantEmv.POIEmvCoreManager.EmvDrlConstraints.
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.18 Set RuPay terminal parameters

Prototype	int setService(android.os.Bundle bundle)
Function	Set RuPay terminal parameters
Parameters	bundle: Parameter values. Enumerated values are as follows: Bundle_Key: ConstantEmv.POIEmvCoreManager.EmvServiceConstraints.CONFIG Bundle_value: ByteArray in TLV format. For specific parameter values, refer to ConstantEmv.POIEmvCoreManager.EmvServiceConstraints.
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.19 Delete RuPay terminal parameters

Prototype	int deleteService()
Function	Delete RuPay payment terminal parameters
Parameters	None
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.20 Retrieve RuPay terminal parameters

Prototype	int getService(android.os.Bundle bundle)
Function	Retrieve RuPay payment terminal parameters
Parameters	bundle: Parameter values. For details, refer to ConstantEmv.POIEmvCoreManager.EmvServiceConstraints.

Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.21 Set Apple VAS transaction parameters

Prototype	int setAppleTerminal(android.os.Bundle bundle)
Function	Set Apple VAS transaction parameters
Parameters	bundle: Parameter values. For details, refer to ConstantEmv.POIEmvCoreManager.AppleTerminalConstraints.
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.22 Retrieve Apple VAS transaction parameters

Prototype	int getAppleTerminal(android.os.Bundle bundle)
Function	Retrieve Apple VAS transaction parameters
Parameters	bundle: Parameter values. For details, refer to ConstantEmv.POIEmvCoreManager.AppleTerminalConstraints.
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.23 Set Apple Merchant transaction parameters

Prototype	int setAppleMerchant(android.os.Bundle bundle)
Function	Set Apple Merchant transaction parameters
Parameters	bundle: Parameter values. For details, refer to ConstantEmv.POIEmvCoreManager.AppleTerminalConstraints.
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.24 Delete Apple Merchant transaction parameters

Prototype	int deleteAppleMerchant()
Function	Delete Apple Merchant transaction parameters
Parameters	None
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.25 Retrieve Apple Merchant transaction parameters

Prototype	int getAppleMerchant(android.os.Bundle bundle)
-----------	--

Function	Retrieve Apple Merchant transaction parameters
Parameters	bundle: Parameter values. For details, refer to <code>ConstantEmv.POIEmvCoreManager.AppleTerminalConstraints</code> .
Return value	0: Success Others: Failure. Refer to <code>EmvError</code> for error codes.
Notes	

3.4.26 Retrieve Kernel version information

Prototype	String getVersion(int type)
Function	Retrieve Kernel version information
Parameters	type: Kernel type. Enumerated values are as follows: <code>ConstantEmv.POIEmvCoreManager.GET_LIB_VERSION</code> <code>ConstantEmv.POIEmvCoreManager.GET_VERSION_EMV</code> <code>ConstantEmv.POIEmvCoreManager.GET_VERSION_VISA</code> <code>ConstantEmv.POIEmvCoreManager.GET_VERSION_MASTERCARD</code> <code>ConstantEmv.POIEmvCoreManager.GET_VERSION_DISCOVER</code> <code>ConstantEmv.POIEmvCoreManager.GET_VERSION_AMEX</code> <code>ConstantEmv.POIEmvCoreManager.GET_VERSION_MIR</code> <code>ConstantEmv.POIEmvCoreManager.GET_VERSION_RUPAY</code> <code>ConstantEmv.POIEmvCoreManager.GET_VERSION_INTERAC</code> <code>ConstantEmv.POIEmvCoreManager.GET_VERSION_APPLE</code>
Return value	Version information
Notes	

3.4.27 Start EMV transaction

Prototype	int startTransaction(android.os.Bundle bundle, IEmvListener callback)
Function	Start EMV transaction
Parameters	bundle: Transaction parameters. For details, refer to <code>ConstantEmv.POIEmvCoreManager.EmvTransDataConstraints</code> . callback: EMV process callback.
Return value	0: Success Others: Failure. Refer to <code>EmvError</code> for error codes.
Notes	

3.4.28 Stop EMV transaction

Prototype	int stopTransaction()
Function	Stop EMV transaction
Parameters	None
Return value	0: Success Others: Failure. Refer to <code>EmvError</code> for error codes.
Notes	

3.4.29 Set EMV Tag/Object parameters

Prototype	int setKernel(byte[] tlv)
Function	Set EMV Tag/Object parameters
Parameters	tlv: TAG in TLV format. Example: HexUtil.parseHex("9F02060000000000119F0306000000000011").
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	After setting, this method is only used to change TAG values during the EMV transaction process. The set TAG values cannot be retrieved using getKernel.

3.4.30 Retrieve EMV Tag/Object parameters

Prototype	byte[] getKernel(String[] tags)
Function	Retrieve EMV Tag/Object parameters
Parameters	tags: Array of TAGs. Example: new String[]{"4F", "50", "87", "9F12"}.
Return value	TAG values in TLV format.
Notes	

3.4.31 Set the result of multiple application selection

Prototype	int setSelectApplicationResponse(int position)
Function	Set the result of multiple application selection
Parameters	position: The selected position in the multiple application selection callback data.
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.32 Set card information confirmation result

Prototype	int setCardInfoResponse(android.os.Bundle bundle)
Function	Set card information confirmation result
Parameters	bundle – parameter value For details, see ConstantEmv.POIEmvCoreManager.EmvCardInfoConstraints
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

3.4.33 Set Pin input result

Prototype	int setPinResponse(android.os.Bundle bundle)
Function	Set Pin input result
Parameters	bundle – parameter value For details, see ConstantEmv.POIEmvCoreManager.EmvPinConstraints
Return value	0: Success Others: Failure. Refer to EmvError for error codes.

Notes	
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3.4.34 Set the online result

Prototype	int setOnlineResponse(android.os.Bundle bundle)
Function	Set the online result
Parameters	bundle – parameter value For details, see ConstantEmv.POIEmvCoreManager.EmvOnlineConstraints
Return value	0: Success Others: Failure. Refer to EmvError for error codes.
Notes	

-- Get general operation module – getGeneralManager --

int setTime(long time)	Set time
long getTime()	Get time
int setTimeZone(String timeZone)	Set time zone
String getTimeZone()	Get time zone
int setBeep(boolean enable, int times, int freq)	Set Beep
int setNavigationBar(int type, boolean isHide)	Set navigation bar
int setStatusbar(boolean isHide)	Set status bar
int wakeUp()	Wake up device
int shutdown()	Shut down device
string getSystemProperty(String key)	Get system property value
setSystemProperty(String key, String value)	Set system property value
int reboot()	Reboot device

3.4.35 Detect card

Prototype	void onEmvProcess(int type, android.os.Bundle info)
Function	Detect card
Parameters	type: Card type. Enumerated values are as follows: ConstantEmv.POIEmvCoreManager.DEVICE_CONTACT ConstantEmv.POIEmvCoreManager.DEVICE_CONTACTLESS ConstantEmv.POIEmvCoreManager.DEVICE_MAGSTRIPE ConstantEmv.POIEmvCoreManager.DEVICE_MIFARE_CLASSIC ConstantEmv.POIEmvCoreManager.DEVICE_MIFARE_ULTRALIGHT ConstantEmv.POIEmvCoreManager.DEVICE_MIFARE_PLUS ConstantEmv.POIEmvCoreManager.DEVICE_MIFARE_DESFIRE info: Card information parameters.

Return value	
Notes	If the transaction card type is a magnetic stripe card, the card data will be returned in this Bundle. For specific parameter constants, refer to ConstantEmv.POIEmvCoreManager.EmvCardInfoConstraints.

3.4.36 Multiple application selection callback

Prototype	void onSelectApplication(List<String> appList, boolean isFirstSelect)
Function	Multiple application selection callback
Parameters	appList: Application selection list. isFirstSelect: Whether it is the first selection.
Return value	
Notes	

3.4.37 Card information confirmation callback

Prototype	void onConfirmCardInfo(int mode, android.os.Bundle info)
Function	Card information confirmation callback
Parameters	mode: Current mode. Enumerated values are as follows: ConstantEmv.POIEmvCoreManager.CMD_TRY_OTHER_APPLICATION ConstantEmv.POIEmvCoreManager.CMD_AMOUNT_CONFIG ConstantEmv.POIEmvCoreManager.CMD_ISSUER_REFERRAL ConstantEmv.POIEmvCoreManager.CMD_GPO_FILTER ConstantEmv.POIEmvCoreManager.CMD_READ_RECORD_FILTER ConstantEmv.POIEmvCoreManager.CMD_SELECT_APPLICATION ConstantEmv.POIEmvCoreManager.CMD_READ_RECORD ConstantEmv.POIEmvCoreManager.CMD_GAC1 ConstantEmv.POIEmvCoreManager.CMD_GAC2 ConstantEmv.POIEmvCoreManager.CMD_SELECT_KERNEL ConstantEmv.POIEmvCoreManager.CMD_SELECT_AFTER ConstantEmv.POIEmvCoreManager.CMD_GPO_BEFORE ConstantEmv.POIEmvCoreManager.CMD_CARD_READ_SUCCESS info: Card information data. For specific parameter constants, refer to ConstantEmv.POIEmvCoreManager.EmvCardInfoConstraints.
Return value	
Notes	

3.4.38 Kernel card scheme type callback

Prototype	void onKernelType(int type)
Function	Kernel card scheme type callback
Parameters	type: Kernel card scheme type. Enumerated values are as follows: ConstantEmv.POIEmvCoreManager.EMV_CARD_NOT ConstantEmv.POIEmvCoreManager.EMV_CARD_VISA

	ConstantEmv.POIEmvCoreManager.EMV_CARD_UNIONPAY ConstantEmv.POIEmvCoreManager.EMV_CARD_MASTERCARD ConstantEmv.POIEmvCoreManager.EMV_CARD_DISCOVER ConstantEmv.POIEmvCoreManager.EMV_CARD_AMEX ConstantEmv.POIEmvCoreManager.EMV_CARD_JCB ConstantEmv.POIEmvCoreManager.EMV_CARD_MIR ConstantEmv.POIEmvCoreManager.EMV_CARD_RUPAY ConstantEmv.POIEmvCoreManager.EMV_CARD_PURE ConstantEmv.POIEmvCoreManager.EMV_CARD_INTERAC ConstantEmv.POIEmvCoreManager.EMV_CARD_EFTPOS
Return value	
Notes	1. This parameter is only returned for contactless card types. 2. For contact transactions and magnetic stripe transactions, this parameter will not be returned.

3.4.39 Second tap card callback

Prototype	void onSecondTapCard()
Function	Second tap card callback
Parameters	
Return value	
Note	

3.4.40 Request PIN input callback

Prototype	void onRequestInputPin(android.os.Bundle info)
Function	Request PIN input callback
Parameters	info: PIN parameters. For details, refer to ConstantEmv.POIEmvCoreManager.EmvPinConstraints.
Return value	
Notes	

3.4.41 Request online process callback

Prototype	void onRequestOnlineProcess(android.os.Bundle info)
Function	Request online process callback
Parameters	info: Online parameters. For details, refer to ConstantEmv.POIEmvCoreManager.EmvOnlineConstraints.
Return value	
Notes	

3.4.42 Transaction result callback

Prototype	void onTransactionResult(int resultCode, android.os.Bundle info)
Function	Transaction result callback
Parameters	resultCode: Transaction result code. 0: Success

	Others: Failure. Refer to ConstantEmv.PosEmvErrorCode/EmvError. info: Transaction data. For details, refer to ConstantEmv.POIEmvCoreManager.EmvResultConstraints.
Return value	
Notes	

➤ 3.5 General Operation function

-- Get general operation module - getGeneralManager --

int setTime(long time)	Set time
long getTime()	Get time
int setTimeZone(String timeZone)	Set time zone
String getTimeZone()	Get time zone
int setBeep(boolean enable, int times, int freq)	Set Beep
int setNavigationBar(int type, boolean isHide)	Set navigation bar
int setStatusBar(boolean isHide)	Set status bar
int wakeUp()	Wake up device
int shutdown()	Shut down device
string getSystemProperty(String key)	Get system property value
setSystemProperty(String key, String value)	Set system property value
int reboot()	Reboot device
int setScreenRotation(boolean enable)	Configure screen rotation functionality

3.5.1 Set buzzer

Prototype	int setBeep(boolean enable, int times, int freq)
Function	Set Beep
Parameters	Parameters: enable - begin or close buzzer times - buzzer duration freq - buzzer frequency
Return value	Return: 0: success Others: failure - see GeneralError
Notes	

3.5.2 Set system navigation bar

Prototype	int setNavigationBar(int type, boolean isHide)
Function	Set system navigation bar
Parameters	Parameters: type – the type of navigation bar to be operated 1: back 2: home 3: recent isHide – the state of the navigation bar to be operated true: hidden false: displayed
Return value	Return: 0: success Others: failure – see GeneralError
Notes	

3.5.3 Set system status bar

Prototype	int setStatusBar(boolean isHide)
Function	Set system status bar
Parameters	isHide – the state of the navigation bar to be operated true: hidden false: displayed
Return value	Return: 0: success Others: failure – see GeneralError
Notes	

3.5.4 Wake up device

Prototype	int wakeUp()
Function	Wake up device
Parameters	
Return value	Return: 0: success Others: failure – see GeneralError
Notes	

3.5.5 Set system time

Prototype	int setTime(long time)
Function	Set system time
Parameters	
Return value	Return: 0: success Others: failure – see GeneralError
Notes	

3.5.6 Get time from system

Prototype	int getTime()
Function	Get time from system
Parameters	
Return value	Return: 0: success Others: failure – see GeneralError
Notes	

3.5.7 Get time zone

Prototype	String getTimeZone()
Function	Get time zone
Parameters	
Return value	Returns: Time zone
Notes	

3.5.8 Set time zone

Prototype	int setTimeZone(String timeZone)
Function	Set time zone
Parameters	Parameters: timeZone – time zone id, supports two formats: 1. Region/City 2. GMT Example 1: setTimeZone("Europe/Moscow") Example 2: setTimeZone("GMT+9")
Return value	Return: 0: success Others: failure – see GeneralError
Notes	

3.5.9 Reboot device

Prototype	int reboot()
Function	Reboot device
Parameters	
Return value	Return: 0: success Others: failure – see GeneralError
Notes	

3.5.10 Shut down device

Prototype	int shutdown()
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Function	Shut down device
Parameters	
Return value	Return: 0: success Others: failure – see GeneralError
Notes	

3.5.11 Set system property value

Prototype	int setSystemProperty(String key, String value)
Function	Set system property value
Parameters	Parameters: key – the key of the system property(ro.product.model) value – the value of the system property
Return value	Return: 0: success Others: failure – see GeneralError
Notes	

3.5.12 Get system property value

Prototype	String getSystemProperty(String key)
Function	Get system property value
Parameters	Parameters: key – the key of the system property (ro.product.model)
Return value	Return: Return the value of the system property by string form, or null if it cannot be read
Notes	

3.5.13 Configures screen rotation function

Prototype	int setScreenRotation(boolean enable)
Function	Configures screen rotation function
Parameters	Parameters: enable – true: Enable screen rotation – false: Disable screen rotation
Return value	0: Success Others: failure – see GeneralError
Notes	

➤ 3.6 PINPAD Operation module

-- Get pinpad module – getPinpadManager --

void cancelInputPin()	Cancel PIN entry
void startInputPin(android.os.Bundle params, android.app.Activity parent, PinpadInputCallback callback)	Start Input PIN, take over TP according to the default view
void startInputPin(android.os.Bundle params, android.content.Context context, Map<String,android.view.View> keyViews, PinpadInputCallback callback)	Start Input PIN, take over TP according to the incoming view

3.6.1 Start Input PIN by provided view

Prototype	void startInputPin(android.os.Bundle params, android.content.Context context, Map<String,android.view.View> keyViews, PinpadInputCallback callback)
Function	Start input PIN, take over TP according to the provided view
Parameters	Parameters: params – PIN configuration parameters for detailed information refer to ConstantEnv.POIEnvCoreManager.EnvPinConstraints keyViews – a collection of views, pass in the following format: <pre>Map<String, View> keyViews = new HashMap<>(); keyViews.put(PinViewEnum.BUTTON0.getType(), button0); keyViews.put(PinViewEnum.BUTTON1.getType(), button1); keyViews.put(PinViewEnum.BUTTON2.getType(), button2); keyViews.put(PinViewEnum.BUTTON3.getType(), button3); keyViews.put(PinViewEnum.BUTTON4.getType(), button4); keyViews.put(PinViewEnum.BUTTON5.getType(), button5); keyViews.put(PinViewEnum.BUTTON6.getType(), button6); keyViews.put(PinViewEnum.BUTTON7.getType(), button7); keyViews.put(PinViewEnum.BUTTON8.getType(), button8); keyViews.put(PinViewEnum.BUTTON9.getType(), button9); keyViews.put(PinViewEnum.BUTTON_ENTER.getType(), buttonEnter); keyViews.put(PinViewEnum.BUTTON_CLEAR.getType(), buttonClear); keyViews.put(PinViewEnum.BUTTON_ESC.getType(), buttonEsc);</pre> context – the context object callback – the PIN input callback
Return value	
Notes	take over TP according to the provided view

3.6.2 Start Input PIN by default view

Prototype	void startInputPin(android.os.Bundle params,
-----------	--

	android.app.Activity parent, PinpadInputCallback callback)
Function	Start input PIN, take over TP according to the default view
Parameters	Parameters: params – PIN configuration parameters Detail information refer to ConstantEmv.POIEmvCoreManager.EmvPinConstraints parent – display the default view based on the provided activity callback – the PIN input callback
Return value	
Notes	take over TP according to the default view

3.6.3 Cancel PIN entry

Prototype	void cancelInputPin()
Function	Cancel PIN entry
Parameters	
Return value	
Notes	

-- PIN input callback – PinpadInputCallback --

void onInput(int len, int key)	Key press event
void onPinError(int verifyResult, int pinTryCntOut)	Callback on error
void onPinSuccess(int verifyResult, byte[] pinBlock, String ksn)	Invoked when the user confirms PIN entry

3.6.4 Key press event

Prototype	void onInput(int len, int key)
Function	Key press event
Parameters	Parameters: len – The length of the password that has been entered key – The current key value, uniformly returns *
Return value	
Notes	

3.6.5 Error callback

Prototype	void onPinError(int verifyResult, int pinTryCntOut)
Function	Error callback

Parameters	Parameters: verifyResult – Error code, refer to ConstantEnv.POIEnvCoreManager.EnvPinConstraints pinTryCntOut – Number of PIN retry attempts
Return value	
Notes	

3.6.6 Return when confirm PIN input

Prototype	void onPinSuccess(int verifyResult, byte[] pinBlock, String ksn)
Function	Return when confirm PIN input
Parameters	Parameters: verifyResult – The result of the PIN confirmation, 0 means successful PIN confirmation. pinBlock – The ciphertext of the password calculated by PINPAD. ksn – Will returned when DUKPT type
Return value	
Notes	

➤ 3.7 Printing Operation module

-- Get printing operation module - getPrinterManager --

int addBarcode(BarcodePrintLine barcodeLine)	Add the barcode to the print cache.
int addBitmap(BitmapPrintLine bitmapLine)	Add the bitmap to the print cache.
int addText(TextPrintLine textLine)	Add the text to the print cache.
int addText(List<TextPrintLine> textLines)	Add a line of text with different styles to the print cache.
void close()	Turn off the printer.
int feedPaper(int lines)	Perform the paper feeding action before starting printing. The main purpose is to adjust the paper position of the printer to ensure that the printed content can be output accurately.
int getPrinterStatus()	Get the current status of the printer.
int open()	Turn on the printer.
int setFont(String path)	Set print font.
int setGray(int gray)	Set print grayscale.
int setLineSpace(int line)	Set print line space.
int startPrint(ConstantPrinter.PrintFailurePolicy failurePolicy, IPrintResultCallback callback)	Execute print job to print the buffered content
int startPrint(IPrintResultCallback callback)	Trigger the printing operation to print out the content in the print cache.
int wrapLine(int lines)	Add a blank line to the print cache, which serves the function of line breaking.

3.7.1 Open the printer module

Prototype	int open()
Function	Open the printer module
Parameters	
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in PrinterError and CommonError
Notes	

3.7.2 Get the current status of the printer

Prototype	int getPrinterStatus()
Function	Get the current status of the printer.
Parameters	
Return value	Return: A negative return value: The operation fails; For the specific meaning of the error code, please refer to the definitions in the PrinterError and CommonError. A positive return value: The operation is successfully executed; For the specific meaning of the status code, please refer to the definitions in the ConstantPrint.
Notes	Must be used after the printer has been opened.

3.7.3 Feed paper before starting printing

Prototype	int feedPaper(int lines)
Function	It is used to feed paper before starting the printing process.
Parameters	Parameters: lines – The number of lines to feed (must be an integer greater than 0).
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in PrinterError and CommonError
Notes	The main purpose is to adjust the printer's paper position, ensuring accurate print output. Must be used after the printer has been opened.

3.7.4 Set print grayscale

Prototype	int setGray(int gray)
Function	Set print grayscale
Parameters	Parameters: gray – Default: 1200.

	Smaller values make prints lighter; larger values make prints darker. Adjust in steps of 200 for consistent results.
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in PrinterError and CommonError
Notes	Must be used after the printer has been opened

3.7.5 Set print font

Prototype	int setFont(String path)
Function	Set print font
Parameters	Parameters: path – The path to the font resource package.
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in PrinterError and CommonError
Notes	Must be used after the printer has been opened.

3.7.6 Set print line space

Prototype	int setLineSpace(int line)
Function	Set print line space
Parameters	Parameters: line – An integer greater than 0.
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in PrinterError and CommonError
Notes	Must be used after the printer has been opened.

3.7.7 Add the text to the print cache

Prototype	int addText(TextPrintLine textLine)
Function	Add the text to the print cache
Parameters	Parameters: textLine – TextPrintLine type(used to define the text content and style for printing).
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in PrinterError and CommonError

Notes	Must be used after the printer has been opened.
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3.7.8 Add a line of text with different styles to the print cache

Prototype	int addText(List<TextPrintLine> textLines)
Function	Add a line of text with different styles to the print cache
Parameters	Parameters: textLines – An array of the TextPrintLine type. Recommend to use the TextPrintLineHelper utility class to construct the TextPrintLine array. This can simplify the development process and make the calls clearer and more straightforward.
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in PrinterError and CommonError
Notes	Must be used after the printer has been opened.

3.7.9 Add a blank line to the print cache

Prototype	int wrapLine(int lines)
Function	Add a blank line to the print cache, which serves the function of line breaking
Parameters	Parameters: lines – Number of lines, an integer greater than 0.
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in PrinterError and CommonError
Notes	Must be used after the printer has been opened.

3.7.10 Add the barcode to the print cache

Prototype	int addBarcode(BarcodePrintLine barcodeLine)
Function	Add the barcode to the print cache
Parameters	Parameters: barcodeLine – BarcodePrintLine Type. Used to define the print content, type, size, and style of the barcode.
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in PrinterError and CommonError
Notes	Must be used after the printer has been opened.

3.7.11 Add the bitmap to the print cache

Prototype	int addBitmap(BitmapPrintLine bitmapLine)
Function	Add the bitmap to the print cache
Parameters	Parameters: bitmapLine – BitmapPrintLine Type. Used to define the picture to be printed and its style.
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in PrinterError and CommonError
Notes	Must be used after the printer has been opened.

3.7.12 Trigger the printing operation to print out the content in the print cache

Prototype	int startPrint(ConstantPrinter.PrintFailurePolicy failurePolicy, IPrintResultCallback callback)
Function	Execute print job to print the buffered content
Parameters	Parameters: failurePolicy – Policy for handling failed prints (how to process remaining tasks after a single print failure); callback – Callback for monitoring print status.
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in PrinterError and CommonError
Notes	Must be used after the printer has been opened

3.7.13 Trigger the printing operation to print out the content in the print cache

Prototype	int startPrint(IPrintResultCallback callback)
Function	Trigger the printing operation to print out the content in the print cache
Parameters	Parameters: callback – Monitor the status callbacks of the printing process.
Return value	Return: 0: The operation is successfully executed; Others: The operation fails. For the specific meaning of the error code, please refer to the definitions in PrinterError and CommonError
Notes	Must be used after the printer has been opened.

3.7.14 Turn off the printer

Prototype	void close()
Function	Turn off the printer

Parameters	
Return value	
Notes	Turn off the printer and release related resources after the printing operation is completed.

-- Print result callback - IPrintResultCallback --

void onFinish()	Callback when the printing finishes successfully.
void onError(int error, String msg)	Callback when an error occurs during the printing process.

3.7.15 Callback when the printing finishes successfully

Prototype	void onFinish()
Function	Callback when the printing finishes successfully
Parameters	
Return value	
Notes	Developers can handle the post - printing - success logic in this method, such as releasing resources, updating the interface status, and closing the printer using the close method.

3.7.16 Callback when an error occurs during the printing process

Prototype	void onError(int error,String msg)
Function	Callback when an error occurs during the printing process
Parameters	Parameters: error - Error code. Refer to PrinterError. msg - Error message.
Return value	
Notes	When receiving the PRINTER_ERROR_PRINT error type, you can try closing the printer first, then reopening it and retrying.

➤ 3.8 Security module

-- Get security module - getSecurityManager --

int calcDes(int keyIndex, int mode, byte[] dataIn, byte[] dataOut)	3DES data encryption and decryption
int calcDukpt(int index, int keyMode, int algType, int mode, byte[] dataIn, byte[] aesIv, byte[] dataOut)	Encrypt or decrypt data using DUKPT
int calcMac(int index,	Calculate MAC

int keyType, byte[] iv, byte[] dataIn, byte[] dataOut)	
int calcMacDukpt(int index, int keyType, int mode, byte[] dataIn, byte[] dataOut)	Calculate MAC using DUKPT
int calcRsa(int keyIndex, int mode, byte[] dataIn, byte[] dataOut)	RSA encryption or decryption
int generateRsaKey(int pubKeyIndex, int priKeyIndex, int size)	Generate RSA keys
int getKCV(int keyIndex, int keyType, byte[] kcvOut)	Get the KCV for the corresponding key type
int getKsnDukpt(int index, byte[] ksnOut)	Get the KSN value by specified index
int getRandom(int length, byte[] keyOut)	Get random numbers compliant with NIST SP800-90A
int increaseKsnDukpt(int index)	Increase KSN
int readRsaKey(int keyIndex, byte[] keyOut, byte[] modulusOut, byte[] exponentOut)	Read RSA keys
int writeKey(int srcKeyType, int srcKeyIndex, int keyType, int index, byte[] keyIn, byte[] kcv)	Write keys
int writeKeyDukpt(int index, byte[] keyIn, byte[] ksnIn, byte[] kcv)	Write DUKPT keys
int writeRsaKey(int keyIndex, byte[] modulus, byte[] exponent)	Write RSA keys
int calcMacDukptDes(int tikIndex, int operationMode, int ksnMode, int macAlgorithm, byte[] dataIn, byte[] aesIv,	Calculate MAC using DUKPT_TDES algorithm

byte[] dataOut, byte[] usedKsn)	
int eraseAllKey()	Erase all keys
int writeKeyDukptDes(int tikIndex, int tlkIndex, byte[] keyIn, byte[] ksnIn, byte[] kcv)	Write DUKPT_DES keys
int writeKeyMKSK(int srcKeyType, int srcKeyIndex, int keyType, int encryptionAlgorithm, int index, byte[] keyIn, byte[] kcv)	Inject MK/SK keys

3.8.1 Write keys

Prototype	int writeKey(int srcKeyType, int srcKeyIndex, int keyType, int index, byte[] keyIn, byte[] kcv)
Function	Write keys
Parameters	Parameters: srcKeyType – Source key type Such as ConstantSecurity.PED_TLK, ConstantSecurity.PED_TMK srcKeyIndex – Source key index the key index used to decrypt the target key. TLK: Only 1 group is supported, and the index range is [1,1]. TMK: Supports 64 groups, the index range is [1, 64]. TPK: Supports 64 groups, the index range is [1, 64]. TAK: Supports 64 groups, the index range is [1, 64]. TDK: Supports 64 groups, the index range is [1, 64]. TEK: Supports 64 groups, the index range is [1, 64]. TTK: Supports 64 groups, the index range is [1, 64]. Note: TPK, TAK, TDK, TEK, TTK share the index space. Key indexes can not be duplicated. If duplication occurs, the later written key will overwrite the previously injected key. keyType – Key type, as follows: ConstantSecurity.PED_TLK ConstantSecurity.PED_TMK ConstantSecurity.PED_TPK ConstantSecurity.PED_TAK ConstantSecurity.PED_TDK

	ConstantSecurity.PED_TEK ConstantSecurity.PED_TTK index – Key index, as follows: TLK: Only 1 group is supported, and the index range is [1,1]. TMK: Supports 64 groups, the index range is [1, 64]. TPK: Supports 64 groups, the index range is [1, 64]. TAK: Supports 64 groups, the index range is [1, 64]. TDK: Supports 64 groups, the index range is [1, 64]. TEK: Supports 64 groups, the index range is [1, 64]. TTK: Supports 64 groups, the index range is [1, 64]. Note: TPK, TAK, TDK, TEK, TTK share the index space. Key indexes should not be duplicated. If duplication occurs, the later written key will overwrite the previously injected key. keyIn – Key information, supported key lengths: [8, 16, 24, 32] kcv – KCV value
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to SecurityError for more information
Notes	

3.8.2 Write DUKPT keys

Prototype	int writeKeyDukpt(int index, byte[] keyIn, byte[] ksnIn, byte[] kcv)
Function	Write DUKPT keys
Parameters	Parameters: index – Key index (supported range 1 ~ 10) keyIn – Key data ksnIn – Initialization KSN kcv – KCV value
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to SecurityError for more information
Notes	

3.8.3 Generate RSA keys

Prototype	int generateRsaKey(int pubKeyIndex, int priKeyIndex, int size)
Function	Generate RSA keys
Parameters	Parameters: pubKeyIndex – Public key index

	priKeyIndex – Private key index size – Key size
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to SecurityError for more information
Notes	Note: The key index range is 1 ~ 4, only support two sets of public and private keys.

3.8.4 Write RSA keys

Prototype	int writeRsaKey(int keyIndex, byte[] modulus, byte[] exponent)
Function	Write RSA keys
Parameters	Parameters: keyIndex – Key index modulus – Modulus exponent – Exponent
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to SecurityError for more information
Notes	Note: If the length of the exponent is less than the length of the modulus, the key type written is a public key. If the length of the exponent is equal to the length of the modulus, the key type written is private key.

3.8.5 Read RSA keys

Prototype	int readRsaKey(int keyIndex, byte[] keyOut, byte[] modulusOut, byte[] exponentOut)
Function	Read RSA keys
Parameters	Parameters: keyIndex – Key index keyOut – Key information modulusOut – Key modulus exponentOut – Key exponent
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to SecurityError for more information
Notes	

3.8.6 Get random numbers

Prototype	int getRandom(int length, byte[] keyOut)
Function	Get random numbers compliant with NIST SP800-90A

Parameters	Parameters: length – Length of the random number keyOut – Random number
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to <code>SecurityError</code> for more information
Notes	

3.8.7 Get KCV for the corresponding key type

Prototype	int getKCV(int keyIndex, int keyType, byte[] kcvOut)
Function	Get KCV for the corresponding key type
Parameters	Parameters: keyIndex – Key index keyType – Key type, as follows: ConstantSecurity.PED_TLK ConstantSecurity.PED_TMK ConstantSecurity.PED_TPK ConstantSecurity.PED_TAK ConstantSecurity.PED_TDK ConstantSecurity.PED_TEK ConstantSecurity.PED_TIK ConstantSecurity.PED_TTK kcvOut – KCV value
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to <code>SecurityError</code> for more information
Notes	

3.8.8 Encrypt or decrypt data by DUKPT

Prototype	int calcDukpt(int index, int keyMode, int algType, int mode, byte[] dataIn, byte[] aesIv, byte[] dataOut)
Function	Encrypt or decrypt data by DUKPT
Parameters	Parameters: index – Key index keyMode – Dukpt mode, can be as follows ConstantSecurity.PED_CALC_DUKPT_MODE_DEC: 0x01

	ConstantSecurity.PED_CALC_DUKPT_MODE_ENC: 0x02 algType – Algorithm type, can be as follows ConstantSecurity.KEY_ALG_TYPE_2TDEA ConstantSecurity.KEY_ALG_TYPE_3TDEA ConstantSecurity.KEY_ALG_TYPE_AES_128 ConstantSecurity.KEY_ALG_TYPE_AES_192 ConstantSecurity.KEY_ALG_TYPE_AES_256 mode – Operation Mode, can be as follows ConstantSecurity.DUKPT_MAC_MODE_ECB: ECB Mode. ConstantSecurity.DUKPT_MAC_MODE_CBC: CBC Mode. dataIn – Input data aesIv – Initialization vector, pass null for ECB mode, or pass an 8-byte vector for other encryption modes dataOut – Encrypted data
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to SecurityError for more information
Notes	

3.8.9 RSA encryption or decryption

Prototype	int calcRsa(int keyIndex, int mode, byte[] dataIn, byte[] dataOut)
Function	RSA encryption or decryption
Parameters	Parameters: keyIndex – Key index mode – RSA padding Mode, can be as follows ConstantSecurity.PED_CALC_RSA_MODE_NO_PADDING ConstantSecurity.PED_CALC_RSA_MODE_PKCS1_PADDING ConstantSecurity.PED_CALC_RSA_MODE_OAEP_PADDING dataIn – Input data dataOut – Encrypted or decrypted data
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to SecurityError for more information
Notes	

3.8.10 3DES data encryption and decryption

Prototype	int calcDes(int keyIndex, int mode, byte[] dataIn, byte[] dataOut)
Function	3DES data encryption and decryption

Parameters	Parameters: keyIndex – Index of TDK. mode – Encryption and Decryption Mode, can be as follows ConstantSecurity.PED_CALC_DES_MODE_ECB_DEC ConstantSecurity.PED_CALC_DES_MODE_ECB_ENC ConstantSecurity.PED_CALC_DES_MODE_CBC_DEC ConstantSecurity.PED_CALC_DES_MODE_CBC_ENC dataIn – Input data dataOut – Encrypted or decrypted data
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to SecurityError for more information
Notes	

3.8.11 Calculate MAC

Prototype	int calcMac(int index, int keyType, byte[] iv, byte[] dataIn, byte[] dataOut)
Function	Calculate MAC
Parameters	Parameters: index – TAK key index keyType – MAC algorithm type, can be as follows ConstantSecurity.MAC_MODE_CBC: CBC-MAC. ConstantSecurity.MAC_MODE_XOR_ECB_MAC: XOR-ECB-MAC. ConstantSecurity.MAC_MODE_ANSI_X9_19: ANSI-X9.19 MAC. ConstantSecurity.MAC_MODE_ANSI_X9_9: ANSI-X9.9 MAC. iv – Initialization vector dataIn – input data dataOut – Calculated data
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to SecurityError for more information
Notes	

3.8.12 Calculate MAC by DUKPT

Prototype	int calcMacDukpt(int index, int keyType, int mode, byte[] dataIn, byte[] dataOut)
Function	Calculate MAC by DUKPT

Parameters	index – dukpt key index keyType – Key Type : This Code is Coded by Two Part: X Y E.g 0x00 0x40. X (Key Usage) can be as follows 0x00 : ConstantSecurity.AUTHENTICATION_GENERATION. 0x01 : ConstantSecurity.AUTHENTICATION_VERIFICATION. 0x02 : ConstantSecurity.AUTHENTICATION_BOTH. Y (Derive Key Algorithm Type) can be as follows 0x00 : ConstantSecurity.KEY_ALG_TYPE_2TDEA. 0x10: ConstantSecurity.KEY_ALG_TYPE_3TDEA. 0x20: ConstantSecurity.KEY_ALG_TYPE_AES_128. 0x30: ConstantSecurity.KEY_ALG_TYPE_AES_192. 0x40: ConstantSecurity.KEY_ALG_TYPE_AES_256. mode – MAC Operation Control Code. This Code is Coded by Three Part: X Y Z. E.g 0x00 0x04 0x80. X (Algorithm Type) can be as follows 0x00 : ConstantSecurity.MAC_MODE_CBC. 0x01 : ConstantSecurity.MAC_MODE_XOR_ECB_MAC. 0x02 : ConstantSecurity.MAC_MODE_ANSI_X9_19. Y (KSN Self Increasing Mode) can be as follows 0x00 : ConstantSecurity.NOT_SELF_INCREASING. 0x40: ConstantSecurity.SELF_INCREASING. Z (Dukpt Mode,default is 0x80) can be as follows 0x80 : ConstantSecurity.DUKPT_MODE_AES_MODE. dataIn – data involved in the calculation dataOut – data after calculation
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to SecurityError for more information
Notes	

3.8.13 Get the KSN for the key at the specified index

Prototype	int getKsnDukpt(int index, byte[] ksnOut)
Function	Get the KSN for the key at the specified index
Parameters	Parameters: index – Key index ksnOut – KSN value
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to SecurityError for more information
Notes	

3.8.14 Increase KSN

Prototype	int increaseKsnDukpt(int index)
Function	Increase KSN
Parameters	Parameters: index – Key index
Return value	Returns: 0: The operation is successfully executed; Others: The operation fails. Please refer to SecurityError for more information
Notes	

3.8.15 Calculate MAC using the DUKPT_TDES algorithm

Prototype	int calcMacDukptDes(int tikIndex, int operationMode, int ksnMode, int macAlgorithm, byte[] dataIn, byte[] aesIv, byte[] dataOut, byte[] usedKsn)
Function	Calculate MAC using the DUKPT_TDES algorithm
Parameters	Parameters: tikIndex – DUKPT key index operationMode – Encryption mode: • OPERATION_MODE_ECB • OPERATION_MODE_CBC ksnMode – KSN mode: • KSN_AUTO_INCREASING_BY_DUKPT_TDES_MAC_BOTH_KEY • KSN_NOT_AUTO_INCREASING_BY_DUKPT_TDES_MAC_BOTH_KEY • KSN_NOT_AUTO_INCREASING_BY_DUKPT_TDES_MAC_RSP_KEY macAlgorithm – MAC algorithm: • MAC_ALGORITHM_CBC • MAC_ALGORITHM_XOR_ECB_MAC • MAC_ALGORITHM_ANSI_X9_19 (TDES only) • MAC_ALGORITHM_ANSI_X9_9 (TDES only) dataIn – Input data to be used in MAC calculation aesIv – Initialization vector (8 bytes for CBC; null for ECB) dataOut – Output MAC result (8 bytes) usedKsn – KSN used in calculation
Return Value	Return: 0 – Success; Others – Failure. See SecurityError for details
Notes	This method calculates a MAC value using the DUKPT TDES algorithm with the provided input, mode, and algorithm settings.

3.8.16 Erase all keys

Prototype	int eraseAllKey()
Function	Erase all keys
Parameters	
Return Value	Return: 0 – Success; Others – Failure. See SecurityError for details
Notes	This method deletes all stored cryptographic keys.

3.8.17 Write DUKPT_DES key

Prototype	int writeKeyDukptDes(int tikIndex, int tlkIndex, byte[] keyIn, byte[] ksnIn, byte[] kcv)
Function	Write DUKPT_DES key
Parameters	Parameters: tikIndex – TIK key index (valid range: 1 – 10) tlkIndex – TLK key index (set to 0 if TLK is not used) keyIn – Key data ksnIn – Initial KSN kcv – Key Check Value
Return Value	Return: 0 – Success; Others – Failure. See SecurityError for details
Notes	This method writes a DUKPT DES key along with its initial KSN and KCV to the specified index.

3.8.18 Write MK/SK key

Prototype	int writeKeyMKSK(int srcKeyType, int srcKeyIndex, int keyType, int encryptionAlgorithm, int index, byte[] keyIn, byte[] kcv)
Function	Write MK/SK key
Parameters	Parameters: srcKeyType – Source key type: • PED_TLK • PED_TMK srcKeyIndex – Source key index: • TLK: only index 1 is supported • TMK: index range [1, 64]

	keyType – Target key type: • PED_TLK, PED_TMK, PED_TPK, PED_TAK, PED_TDK, PED_TEK, PED_TTK encryptionAlgorithm – Encryption algorithm: • ENCRYPTION_ALGORITHM_TDES, ENCRYPTION_ALGORITHM_AES index – Key index: • TLK: [1,1]; others (TMK, TPK, etc.): [1,64] Note: TPK, TAK, TDK, TEK, TTK share the same index space; avoid duplication to prevent overwriting. keyIn – Key data (supported lengths: 8, 16, 24, 32 bytes) kcv – Key check value
Return Value	Return: 0 – Success; Others – Failure. See SecurityError for details
Notes	This method writes Master or Session keys (MK/SK). Not applicable for DUKPT key injection.

➤ 3.9 ECR module

-- Get ECR module – getEcrManager --

int close(String port)	Close the connection on the specified port; pass 'null' to close all ports
ArrayList<String> getLocal()	Enumerate and return available port numbers
EcrConnector openOrConnect(String connection)	Create a connection; Supported types: Serial, USB-to-Serial, Cash Drawer

3.9.1 Create a connection to a device

Prototype	EcrConnector openOrConnect(String connection)
Function	Create a connection to a device
Parameters	Parameters: connection – Connection parameters in the format: Serial/USB/Cash Drawer: "PORT,BAUD" Bluetooth: "UUID,BT" WiFi: "URL,WIFI"
Return Value	Return: EcrConnector – Operation object
Notes	Supports multiple connection types including serial, USB-serial, cash drawer, Bluetooth, and WiFi.

3.9.2 Close connection(s)

Prototype	int close(String port)
Function	Close connection(s)
Parameters	Parameters: port – Port name to close, or 'null' to close all ports

Return Value	Return: 0 – Success; Others – Failure
Notes	Closes the specified port or all connections.

3.9.3 Enumerate and return available ports

Prototype	ArrayList<String> getLocal()
Function	Enumerate and return available ports
Parameters	
Return Value	Return: List of local port names (e.g., "tty0" for /dev/ttySx)
Notes	Used to detect available local communication ports.

-- ECR connector object – EcrConnector --

String getConnectInfo()	Return connection information
byte[] read()	Read data
void write(byte[] commands)	Write data

3.9.8 Read data

Prototype	byte[] read()
Function	Read data
Parameters	
Return Value	Return: Byte array containing the read data
Notes	Reads data from the scanner module or connected device.

3.9.9 Write data

Prototype	void write(byte[] commands)
Function	Write data
Parameters	Parameters: commands – Command data to write (max 4096 bytes)
Return Value	
Notes	Sends command data to the connected device or module.

3.9.10 Get connection information

Prototype	String getConnectInfo()
Function	Get connection information
Parameters	
Return Value	Return: Connection information string
Notes	Returns current connection details of the scanner or related module.

➤ 3.10 Scanner module

-- Get scanner module - getScannerManager --

void close()	Close the scanner module
boolean isBarcodeEnabled(ConstantScanner.BarcodeFormat type)	Check whether a specific barcode format is enabled
void open(ConstantScanner.ScannerCameraType cameraType, IConnectionStatusListener callback)	Open the scanner module with a specific camera type
void open(IConnectionStatusListener callback)	Open the scanner module
int registerResultCallback(IScannerResultCallback callback)	Register a callback to receive scan results
int setBarcodeEnable(boolean enable)	Enable/disable all supported barcode formats
int setBarcodeEnable(List<ConstantScanner.BarcodeFormat> types, boolean enable)	Enable/disable specified barcode formats
int startScan()	Start scanning
int stopScan()	Stop scanning
int switchLight()	Toggle the fill light on or off

3.10.1 Open the scanner module

Prototype	void open(IConnectionStatusListener callback)
Function	Open the scanner module
Parameters	Parameters: callback - Connection status callback
Return Value	
Notes	Initializes the scanner module. Other scanner interfaces should only be used after a successful initialization.

3.10.2 Open the scanner module with a specific camera

Prototype	void open(ConstantScanner.ScannerCameraType cameraType, IConnectionStatusListener callback)
Function	Open the scanner module with a specific camera
Parameters	Parameters: cameraType - Camera type for scanning: 1. CAMERA_REAR, 2. CAMERA_FRONT, 3. SCANNER callback - Connection status callback
Return Value	
Notes	Initializes the scanner with the selected camera. Other interfaces should only be used after successful initialization.

3.10.3 Close the scanner module

Prototype	void close()
Function	Close the scanner module
Parameters	
Return Value	
Notes	Unbinds from the scanner service and releases related resources.

3.10.4 Register a scanner result callback

Prototype	int registerResultCallback(IScannerResultCallback callback)
Function	Register a scanner result callback
Parameters	Parameters: callback – Callback for scan results including type and data
Return Value	Return: 0 – Success; Others – Failure. See ScannerError and CommonError for details
Notes	Must be called after open() is successfully executed.

3.10.5 Start scanning

Prototype	int startScan()
Function	Start scanning
Parameters	
Return Value	Return:0 – Success; Others – Failure. See ScannerError and CommonError for details
Notes	Triggers a scan. Must be called after open() is successfully executed.

3.10.6 Stop scanning

Prototype	int stopScan()
Function	Stop scanning
Parameters	
Return Value	Return:0 – Success; Others – Failure. See ScannerError and CommonError for details
Notes	Stops the current scan. Must be called after open() is successfully executed.

3.10.7 Enable or disable specific barcode types

Prototype	int setBarcodeEnable(List<ConstantScanner.BarcodeFormat> types, boolean enable)
Function	Enable or disable specific barcode types
Parameters	Parameters: types – List of barcode formats to enable/disable enable – true to enable, false to disable
Return Value	Return:

	0 – Success; Others – Failure. See ScannerError and CommonError for details
Notes	Must be called after open() is successfully executed.

3.10.8 Enable or disable all barcode types

Prototype	int setBarcodeEnable(boolean enable)
Function	Enable or disable all barcode types
Parameters	Parameters: Enable – true to enable all types; false to disable all types
Return Value	Return: 0 – Success; Others – Failure. See ScannerError and CommonError for details
Notes	Enable or disable all supported code types. This method must be called after a successful open().

3.10.9 Check if a barcode type is enabled

Prototype	boolean isBarcodeEnabled(ConstantScanner.BarcodeFormat type)
Function	Check if a barcode type is enabled
Parameters	Parameters: type – The barcode format to check
Return Value	Return: true to enabled; false to disabled or SDK error
Notes	Must be called after open() is successfully executed.

3.10.10 Toggle the flashlight

Prototype	int switchLight()
Function	Toggle the flashlight
Parameters	
Return Value	Return: 0 – Success; Others – Failure. See ScannerError and CommonError for details
Notes	Requires hardware support; Not supported by SCANNER Must be called after open() is successfully executed.

-- Scanner result callback – ScannerResultCallback --

void onResult(String sym, String barcode)	Retrieve the scan result and the barcode format
--	--

3.10.11 Receive scan result and symbology type

Prototype	void onResult(String sym, String barcode)
Function	Receive scan result and symbology type

Parameters	Parameters: sym – Barcode symbology type barcode – Scanned barcode result
Return Value	
Notes	This method is triggered upon a successful scan, returning the barcode data and type.

-- Connection Status Listener- IConnectionStatusListener--

void onConnected()	Called when the connection to the scanning service is established
void onDisconnected()	Called when the connection to the scanning service is lost
void onError(int error, String msg)	Called when the connection to the scanning service fails

3.10.12 Called when the connection to the scanner service is established

Prototype	void onConnected()
Function	Called when the connection to the scanner service is established
Parameters	
Return Value	
Notes	Indicates that the scanner module has been successfully initialized.

3.10.13 Called when the connection to the scanner service fails

Prototype	void onError(int error, String msg)
Function	Called when the connection to the scanner service fails
Parameters	Parameters: error – Error code. See ScannerError and CommonError for definitions msg – Error description
Return Value	
Notes	Triggered upon failure to establish a connection with the scanner service.

3.10.14 Called when the connection to the scanner service is lost

Prototype	void onDisconnected()
Function	Called when the connection to the scanner service is lost
Parameters	
Return Value	
Notes	Indicates that the scanner service is no longer connected.

● 4. Error Code Definition

4.1 CardReaderError

Error Code	Error Description	Error Value
CARD_APDU_OTHER_ERROR	APDU Error	-70004
CARD_CHECK_OTHER_ERROR	Card Check Other Error	-70003
CARD_CONTACT_ATR_ERROR	Contact Card ATR Data Error	-72001
CARD_CONTACT_NO_SUPPORT	IC Card Not Supported	-72000
CARD_CONTACT_OTHERS_ERROR	Contact Card Insertion Failed	-72002
CARD_CONTACTLESS_ATS_ERROR	Contactless Card ATS Data Error	-73001
CARD_CONTACTLESS_NO_SUPPORT	IC Card Not Supported	-73000
CARD_CONTACTLESS_OTHERS_ERROR	Contactless Card Other Error	-73002
CARD_EXIST_STATUS_OTHER_ERROR	Card Present Status Other Error	-70005
CARD_FELICA_ATS_ERROR	Contactless Card ATS Data Error	-74001
CARD_FELICA_NO_SUPPORT	IC Card Not Supported	-74000
CARD_FELICA_OTHERS_ERROR	Contactless Card Other Error	-74002
CARD_MAG_INVALID_ERROR	Magnetic Track or Data Verification Failed	-71003
CARD_MAG_NEED_RETRY_ERROR	Magnetic Card Swipe Failed, Please Retry	-71002
CARD_MAG_NO_DATA_ERROR	No Data in Magnetic Track	-71001
CARD_MAG_NO_SUPPORT	Magnetic Card Not Supported	-71000
CARD_MAG_OTHERS_ERROR	Magnetic Card Swipe Failed	-71004
CARD_READER_CONTACT_ALREADY_CLOSE	IC Card Already Powered Off	-72006
CARD_READER_CONTACT_CHECK_ERROR	Contact Card Check Failed	-72004
CARD_READER_CONTACT_OFF_ERROR	Contact Card Power-Off Failed	-72005
CARD_READER_CONTACT_OPEN_ERROR	Magnetic Card Check Open Failed	-72003
CARD_READER_CONTACTLESS_ALREADY_CLOSE	Contactless Card Already Powered Off	-73006
CARD_READER_CONTACTLESS_CHECK_ERROR	Contactless Card Check Failed	-73004
CARD_READER_CONTACTLESS_MULTI_CARD	Multiple Contactless Cards Detected	-73007
CARD_READER_CONTACTLESS_OFF_ERROR	Contactless Card Power-Off Failed	-73005
CARD_READER_CONTACTLESS_OPEN_ERROR	Magnetic Card Check Open Failed	-73003
CARD_READER_FELICA_ALREADY_CLOSE	Contactless Card Already Powered Off	-74006
CARD_READER_FELICA_CHECK_ERROR	Contactless Card Check Failed	-74004
CARD_READER_FELICA_OFF_ERROR	Contactless Card Power-Off Failed	-74005
CARD_READER_FELICA_OPEN_ERROR	Magnetic Card Check Open Failed	-74003
CARD_READER_MAG_CHECK_ERROR	Magnetic Card Check Failed	-71006
CARD_READER_MAG_OFF_ERROR	Magnetic Card Power-Off Failed	-71007
CARD_READER_MAG_OPEN_ERROR	Magnetic Card Check Open Failed	-71005
CARD_READER_OFF_OTHER_ERROR	Card Power-Off Other Error	-70002
CARD_TYPE_ERROR	Card Type Error	-70001
NFC_TAG_ALREADY_CLOSE	NFC TAG already closed	-75006
NFC_TAG_NDEF_MESSAGE_ERROR	NFC TAG data error	-75004
NFC_TAG_NDEF_MESSAGE_LENGTH_ERROR	NDEFMessage data length exceeds limit	-75005

	(max: 255 bytes)	
NFC_TAG_NO_SUPPORT	NFC TAG not supported	-75000
NFC_TAG_OFF_ERROR	NFC TAG close failed	-75002
NFC_TAG_OPEN_ERROR	NFC TAG open failed	-75001
NFC_TAG_OTHERS_ERROR	Other NFC TAG error	-75007
NFC_TAG_WRITE_MESSAGE_ERROR	NFC TAG write failed	-75003
NOT_SUPPORT_CHECK_CONTACTLESS_AND_FELICA_SIMULTANEOUS	Simultaneous contactless and Felica card check not supported	-73008
NOT_SUPPORT_OPEN_CONTACTLESS_AND_FELICA_SIMULTANEOUS	Simultaneous contactless and Felica card power-on not supported	-73009

4.2 CommonError

Error Code	Error Description	Error Value
FINANCIAL_PARAMETERS_INVALID	Illegal parameters	-10002
FINANCIAL_SERVICE_DISCONNECT	Financial services are not connected, please initialize financial services	-10001
FINANCIAL_VERSION_NOT_MATCH	Financial Services SDK version mismatch	-10000

4.3 GeneralError

Error Code	Error Description	Error Value
GENERAL_ERROR_INIT	Basic operation module is not initialized	-20000
GENERAL_NAVIGATION_BUTTON_TYPE_INVALID	Navigation bar button type is invalid; only back, home, and recent buttons are supported by default	-20003
GENERAL_OTHER_ERROR	Other exceptions, e.g., system interface call errors	-20001
GENERAL_PARAMETERS_INVALID	Invalid parameters	-20002
GENERAL_SCREEN_ROTATION_ERROR	Screen rotation anomaly configuration	-20004

4.4 PinpadError

Error Code	Error Description	Error Value
PINPAD_CANCEL_ERROR	Other Errors	-60002
PINPAD_OTHER_ERROR	Other Errors	-60000
PINPAD_SCREEN_ORIENTATION_CHANGED	Screen rotation	-60003
PINPAD_START_ERROR	Other Errors	-60001

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4.5 PrinterError

Error Code	Error Description	Error Value
PRINTER_ERROR_INIT	Printer not initialized or initialization failed	-50000
PRINTER_ERROR_LOW_POWER	System power is below the low power threshold	-50006
PRINTER_ERROR_NO_CONTENT	No print content added	-50007
PRINTER_ERROR_NO_PAPER	Printer out of paper	-50005
PRINTER_ERROR_NO_PRINTER	No printing device found	-50001
PRINTER_ERROR_NOT_OPENED	Printer not opened	-50002
PRINTER_ERROR_OTHER	Other unknown errors	-50008
PRINTER_ERROR_OVERHEAT	Printer overheating	-50004
PRINTER_ERROR_PRINT	Printing failed	-50003
PRINTER_ERROR_QUEUE_OVER_FLOW	Print Job Buffer Overflow	-50009
PRINTER_ERROR_SCREEN_OFF	Printing Disabled in Sleep Mode	-50010

4.6 ScannerError

Error Code	Error Description	Error Value
SCANNER_OTHER_ERROR	Other unknown errors	-30004
SCANNER_SERVICE_CONNECT_FAILED	Failed to connect to the scanning service	-30002
SCANNER_SERVICE_DISCONNECT	The scanning service is not connected, please call the open method to initialize it first	-30003
SCANNER_SERVICE_UNAVAILABLE	The scanning service is unavailable	-30001
SCANNER_SERVICE_NOT_FIND	The scanning service was not found	-30000

4.7 SecurityError

Error Code	Error Description	Error Value
SECURITY_ERROR_INIT	Security algorithm module is not initialized	-40000
SECURITY_OTHER_ERROR	Other error, e.g., system interface call error	-40001
SECURITY_PARAMETERS_INVALID	Illegal parameters, e.g., parameters are empty or in an invalid format.	-40002
SECURITY_KEY_INDEX_OUT_OF_RANGE	Key index out of range	-40003

4.8 EmvError

Error Code	Error Description	Error Value
EMV_AID_OTHER_ERROR	Other AID parameter error	-80100

EMV_APPLE_TERMINAL_OTHER_ERROR	Other Apple Terminal configuration error	-80400
EMV_CAPK_OTHER_ERROR	Other CAPK error	-80200
EMV_DELETE_AID_ERROR	Error deleting AID parameter	-80109
EMV_DELETE_APPLE_MERCHANT_ERROR	Error deleting Apple Merchant	-80405
EMV_DELETE_APPLE_TERMINAL_ERROR	Error deleting Apple Terminal	-80402
EMV_DELETE_CAPK_ERROR	Error deleting CAPK	-80209
EMV_DELETE_DRL_ERROR	Error deleting DRL configuration	-80310
EMV_DELETE_EXCEPTION_FILE_ERROR	Error deleting ExceptionFile	-80305
EMV_DELETE_REVOCATION_IPK_ERROR	Error deleting RevocationIPK	-80308
EMV_DELETE_SERVICE_ERROR	Error deleting RuPay Service	-80314
EMV_DRL_OTHER_ERROR	Other DRL configuration error	-80309
EMV_GET_APPLE_MERCHANT_ERROR	Error retrieving Apple Merchant	-80404
EMV_GET_DRL_ERROR	Error retrieving DRL configuration	-80311
EMV_GET_KERNEL_VERSION_ERROR	Error retrieving Kernel version	-80316
EMV_GET_SERVICE_ERROR	Error retrieving RuPay Service	-80315
EMV_GET_TERMINAL_ERROR	Error retrieving terminal parameters	-80302
EMV_OTHER_ERROR	Other EMV error	-80001
EMV_SERVICE_OTHER_ERROR	Other RuPay Service error	-80312
EMV_SET_AID_ERROR	Error setting AID parameter	-80108
EMV_SET_AID_ERROR_AID_NULL	Error loading AID parameter, AID is null	-80101
EMV_SET_AID_ERROR_DDOL_NULL	Error loading AID parameter, DDOL is null	-80103
EMV_SET_AID_ERROR_TACDEFAULT_NULL	Error loading AID parameter, TACDefault is null	-80107
EMV_SET_AID_ERROR_TACDENIAL_NULL	Error loading AID parameter, TACDenial is null	-80105
EMV_SET_AID_ERROR_TACONLINE_NULL	Error loading AID parameter, TACOnline is null	-80106
EMV_SET_AID_ERROR_TDOL_NULL	Error loading AID parameter, TDOL is null	-80104
EMV_SET_AID_ERROR_VERSION_NULL	Error loading AID parameter, Version is null	-80102
EMV_SET_APPLE_MERCHANT_ERROR	Error setting Apple Merchant	-80403
EMV_SET_APPLE_TERMINAL_ERROR	Error setting Apple Terminal	-80401
EMV_SET_CAPK_ERROR	Error setting CAPK	-80208
EMV_SET_CAPK_ERROR_ALGORITHM_IND_NULL	Error loading CAPK parameter, AlgorithmInd is null	-80206
EMV_SET_CAPK_ERROR_CAPK_INDEX_NULL	Error loading CAPK parameter, CapkIndex is null	-80202
EMV_SET_CAPK_ERROR_CHECKSUM_NULL	Error loading CAPK parameter, Checksum is null	-80205
EMV_SET_CAPK_ERROR_Exponent_NULL	Error loading CAPK parameter,	-80204

	Exponent is null	
EMV_SET_CAPK_ERROR_HASH_IND_NULL	Error loading CAPK parameter, HashInd is null	-80207
EMV_SET_CAPK_ERROR_MODULE_NULL	Error loading CAPK parameter, Module is null	-80203
EMV_SET_CAPK_ERROR_RID_NULL	Error loading CAPK parameter, RID is null	-80201
EMV_SET_CARD_INFO_RESPONSE_ERROR	Error setting card confirmation result	-81004
EMV_SET_DRL_ERROR	Error setting DRL configuration	-80309
EMV_SET_EXCEPTION_FILE_ERROR	Error setting ExceptionFile	-80304
EMV_SET_KERNEL_ERROR	Error setting KernelTag	-80317
EMV_SET_ONLINE_RESPONSE_ERROR	Error setting online result	-81006
EMV_SET_PIN_RESPONSE_ERROR	Error setting PIN result	-81005
EMV_SET_REVOCATION_IPK_ERROR	Error setting RevocationIPK	-80307
EMV_SET_SELECT_RESPONSE_ERROR	Error setting multi-application selection result	-81003
EMV_SET_SELECT_RESPONSE_POSITION_ERROR	Error confirming position in multi-application selection result	-81007
EMV_SET_SERVICE_ERROR	Error setting RuPay Service	-80313
EMV_SET_TERMINAL_ERROR	Error loading terminal parameters	-80301
EMV_START_TRANSACTION_ERROR	Error starting transaction	-81001
EMV_STOP_TRANSACTION_ERROR	Error stopping transaction	-81002
EMV_SUCCESS	Success	0
EMV_TERMINAL_EXCEPTION_FILE_OTHER_ERROR	Other ExceptionFile error	-80303
EMV_TERMINAL_OTHER_ERROR	Other terminal parameter error	-80300
EMV_TERMINAL_REVOCATION_IPK_OTHER_ERROR	Other RevocationIPK error	-80306

4.9 PosEmvErrorCode

Error Code	Error Description	Error Value
APPLE_VAS_APPROVED	Apple Vas Approved.	6
APPLE_VAS_FAILED	Apple Vas Failed.	-41
APPLE_VAS_UNTREATED	Apple Vas Untreated.	-40
APPLE_VAS_WAITING_ACTIVATION	Apple Vas Waiting Activation.	-43
APPLE_VAS_WAITING_INTERVENTION	Apple Vas Waiting Intervention.	-42
EMV_APP_BLOCKED	Application Blocked.	-7
EMV_APP_EMPTY	No Application.	-9
EMV_APPROVED	Transaction Approved.	1
EMV_APPROVED_ONLINE	Transaction Online Approved.	2
EMV_CANCEL	Trans Cancel.	-1
EMV_CARD_BLOCKED	Card Locked.	-8
EMV_CARD_ERROR	Abecs Fallback.	-40
EMV_COMMAND_FAIL	Read the Card Fail.	-3
EMV_DATA_ERROR	Card Data Error.	-81

EMV_DECLINED	Transaction Declined.	3
EMV_DELAYED_APPROVED	Transaction Delayed Approved.	5
EMV_ENCRYPT_ERROR	Trade Encrypt Error.	-30
EMV_FALLBACK	Fallback.	-4
EMV_FORCE_APPROVED	Transaction Force Approved.	4
EMV_GPO_6985	Command GPO 6985.	-50
EMV_MULTICONTACTLESS	Read Multi Contactless.	-5
EMV_NOT_ACCEPTED	Not Accepted.	-11
EMV_NOT_ALLOWED	Not Allowed.	-10
EMV_OK	Handle OK.	0
EMV_OTHER_ERROR	Other Error Exceptions.	-999
EMV_OTHER_ICC_INTERFACE	Not Pure Magnetic Strip.	-6
EMV_OTHER_INTERFACE	Try Another Interface.	-14
EMV_READ_RECORD_ERROR	Command Read Record Error.	-80
EMV_SEE_PHONE	See Phone.	-13
EMV_TERMINATED	Terminated.	-12
EMV_TIMEOUT	Trans Timeout.	-2
EMV_UNENCRYPTED	Trade Unencrypted.	-31
EXCEPTION_ERROR	Exception Error.	255
PARAMETER_ERROR	Parameter Error.	254

4.10 EcrError

Error Code	Error Description	Error Value
ECR_CONNECT_CLOSE_FAILED	Failed to close connection	-90006
ECR_CONNECT_FAILED	Connection failed	-90003
ECR_CONNECT_INFO_EMPTY	Connection parameter info is empty	-90001
ECR_CONNECT_INFO_FORMAT_EXCEPTION	Connection parameter format exception	-90002
ECR_ERROR_INIT	ECR module not initialized	-90000
ECR_FD_EMPTY	File handle is null	-90004
ECR_FD_FORMAT_EXCEPTION	File handle format exception	-90005
ECR_WRITE_DATA_EMPTY	Write data is empty	-90007
ECR_WRITE_DATA_FAILED	Write data exception	-90008

● 5. Entity Class Definition

5.1 com.kozen.financial.aidl.emv.EmvCapk

Constant Name	Type	Value
ALGO_IND_RSA	int	1
ALGO_IND_SM	int	4
HASH_IND_NOT	int	0
HASH_IND_SHA1	int	1

5.2 ConstantCardReader

Constant Name	Type	Value
ATR	String	"cardAtr"
ATS	String	"cardAs"
CARD_ATTRIBUTE	String	"cardAttribute"
CARD_CATEGORY	String	"cardCategory"
CARD_CHANNEL	String	"cardChannel"
CARD_SERIAL_NUM	String	"cardSerialNum"
CARD_TYPE	String	"cardType"
EXPIRED_DATE	String	"cardExpDate"
ID_FOR_MANUFACTURER	String	"IDm"
PAN	String	"cardPan"
PARAMETER_FOR_MANUFACTURER	String	"PMm"
REQUEST_DATA	String	"requestData"
SERVICE_CODE	String	"cardServiceCode"
TIMEOUT	String	"cardTimeout"
TRACK1	String	"cardTrack1"
TRACK2	String	"cardTrack2"
TRACK3	String	"cardTrack3"

5.3 ConstantCardReader.CardType

Constant Name	Type	Value
ALL	int	0
CONTACT	int	2
CONTACTLESS	int	4
FELICA	int	32
MAGNETIC	int	1
MIFARE	int	8
NFC_TAG	int	64
PSAM	int	16

5.4 ConstantEnv.POIEnvCoreManager

Constant Name	Type	Value
CMD_AMOUNT_CONFIG	int	1
CMD_CARD_READ_SUCCESS	int	64
CMD_GAC1	int	18
CMD_GAC2	int	19
CMD_GPO_BEFORE	int	49
CMD_GPO_FILTER	int	16
CMD_ISSUER_REFERRAL	int	2
CMD_READ_RECORD	int	17
CMD_READ_RECORD_FILTER	int	17

CMD_SELECT_AFTER	int	48
CMD_SELECT_APPLICATION	int	16
CMD_SELECT_KERNEL	int	32
CMD_TRY_OTHER_APPLICATION	int	0
DEVICE_CONTACT	int	1
DEVICE_CONTACTLESS	int	2
DEVICE_MAGSTRIPE	int	4
DEVICE_MIFARE_CLASSIC	int	8
DEVICE_MIFARE_DESFIRE	int	64
DEVICE_MIFARE_PLUS	int	32
DEVICE_MIFARE_ULTRALIGHT	int	16
DEVICE_VICC	int	128
EMV_ADMINISTRATIVE	int	7
EMV_BALANCE_ENQUIRY	int	13
EMV_BALANCE_UPDATE	int	14
EMV_CARD_AMEX	int	5
EMV_CARD_DISCOVER	int	4
EMV_CARD_EFTPOS	int	11
EMV_CARD_INTERAC	int	10
EMV_CARD_JCB	int	6
EMV_CARD_MASTERCARD	int	3
EMV_CARD_MIR	int	7
EMV_CARD_NOT	int	0
EMV_CARD_PURE	int	9
EMV_CARD_RUPAY	int	8
EMV_CARD_UNIONPAY	int	2
EMV_CARD_VISA	int	1
EMV_CASH	int	3
EMV_CASHBACK	int	4
EMV_DEPOSIT	int	10
EMV_DISBURSEMENT	int	8
EMV_GOODS	int	1
EMV_INQUIRY	int	11
EMV_MONEY_ADD	int	12
EMV_PAYMENT	int	6
EMV_REFUND	int	9
EMV_SERVICE	int	2
EMV_SERVICE_CREATION	int	16
EMV_TRANSFER	int	5
EMV_VOID	int	15
GET_LIB_VERSION	int	0
GET_VERSION_AMEX	int	6
GET_VERSION_APPLE	int	16
GET_VERSION_CL1	int	14

GET_VERSION_DISCOVER	int	5
GET_VERSION_EFTPOS	int	12
GET_VERSION_EMV	int	1
GET_VERSION_INTERAC	int	11
GET_VERSION_JCB	int	7
GET_VERSION_L1	int	13
GET_VERSION_MASTERCARD	int	4
GET_VERSION_MIR	int	8
GET_VERSION_PURE	int	10
GET_VERSION_RUPAY	int	9
GET_VERSION_UNIONPAY	int	3
GET_VERSION_VISA	int	2
PIN_ENCIPHER_PIN	int	2
PIN_ONLINE_PIN	int	1
PIN_PLAIN_PIN	int	0

5.5 ConstantEmv.POIEmvCoreManager.AppleTerminalConstraints

Constant Name	Type	Value
CAPABILITY	String	"capability"
CAPABILITY_DUAL_MODE	int	1
CAPABILITY_PAYMENT_ONLY	int	3
CAPABILITY_SINGLE_MODE	int	0
CAPABILITY_VAS_ONLY	int	2
DATA	String	"data"
PROTOCOL	String	"protocol"
PROTOCOL_FULL_VAS	int	1
PROTOCOL_URL_ONLY	int	0
TAG_APPLE_SET_DELIMITER	String	"DF01"
TAG_APPLE_SET_FILTER	String	"9F2B"
TAG_APPLE_SET_MERCHANT_ID	String	"9F25"
TAG_APPLE_SET_MERCHANT_URL	String	"9F29"

5.6 ConstantEmv.POIEmvCoreManager.EmvCardInfoConstraints

Constant Name	Type	Value
ATR	String	"atr"
CARD	String	"card"
DATA	String	"data"
OUT_AMOUNT	String	"amount"
OUT_AMOUNT_OTHER	String	"amountOther"
OUT_CONFIRM	String	"confirm"
OUT_TLV	String	"tlv"
OUT_TVR	String	"tvr"
TRACK1	String	"track1"

TRACK2	String	"track2"
TRACK3	String	"track3"

5.7 ConstantEnv.POIEnvCoreManager.EmvDrlConstraints

Constant Name	Type	Value
CONFIG	String	"Config"
TAG_DRL_SET_CVM_REQUIRED_LIMIT	String	"DF24"
TAG_DRL_SET_DELIMITER	String	"DF01"
TAG_DRL_SET_ENTRY_POINT	String	"DF30"
TAG_DRL_SET_FLOOR_LIMIT	String	"DF25"
TAG_DRL_SET_PROGRAM_ID	String	"9F5A"
TAG_DRL_SET_STATUS_ZERO_AMOUNT	String	"DF32"
TAG_DRL_SET_TRANSACTION_LIMIT	String	"DF23"
TYPE_AMEX	int	2
TYPE_VISA	int	1

5.7 ConstantEnv.POIEnvCoreManager.EmvOnlineConstraints

Constant Name	Type	Value
APPLE_DATA	String	"appleData"
APPLE_MERCHANT	String	"appleMerchant"
APPLE_RESULT	String	"appleResult"
EMV_DATA	String	"emvData"
EMV_ONLINE_APPROVE	int	0
EMV_ONLINE_DENIAL	int	2
EMV_ONLINE_FAIL	int	1
EMV_ONLINE_REFER_TO_CARD_ISSUER	int	3
ENCRYPT_DATA	String	"encryptData"
ENCRYPT_RESULT	String	"encryptResult"
OUT_AUTH_CODE	String	"outAuthCode"
OUT_AUTH_DATA	String	"outAuthData"
OUT_AUTH_RESP_CODE	String	"outAuthRespCode"
OUT_ISSUER_SCRIPT	String	"outIssuerScript"
OUT_SPECIAL_AUTH_RESP_CODE	String	"outSpecialAuthRespCode"

5.8 ConstantEnv.POIEnvCoreManager.EmvPinConstraints

Constant Name	Type	Value
OUT_PIN_BLOCK	String	"outPinBlock"
OUT_PIN_TRY_COUNTER	String	"outPinTryCounter"
OUT_PIN_VERIFY_RESULT	String	"outPinVerifyResult"
PIN_BLOCK_FORMAT	String	"pinBlockFormat"
PIN_BYPASS	String	"pinBypass"
PIN_CARD	String	"pinCard"

PIN_CARD_RANDOM	String	"pinCardRandom"
PIN_COUNTER	String	"pinCounter"
PIN_DUKPT_KEY_LENGTH	String	"pinDukptKeyLength"
PIN_ENCRYPT	String	"pinEncrypt"
PIN_EXPONENT	String	"pinExponent"
PIN_IS_ORDER	String	"isOrder "
PIN_ISO_FMT0	int	0
PIN_ISO_FMT1	int	1
PIN_ISO_FMT1_SM4	int	5
PIN_ISO_FMT2	int	2
PIN_ISO_FMT2_SM4	int	6
PIN_ISO_FMT3	int	3
PIN_ISO_FMT3_SM4	int	7
PIN_ISO_FMT4	int	4
PIN_KEY_INDEX	String	"pinKeyId"
PIN_KEY_MODE	String	"pinKeyMode"
PIN_KEY_MODE_DUKPT	int	3
PIN_KEY_MODE_TPK	int	1
PIN_LENGTH_LIMIT	String	"lengthLimit"
PIN_MODULE	String	"pinModule"
PIN_TIMEOUT	String	"pinTimeout"
PIN_TYPE	String	"pinType"
VERIFY_CANCELED	int	4
VERIFY_ERROR	int	3
VERIFY_NO_PASSWORD	int	1
VERIFY_PIN_BLOCK	int	2
VERIFY_SUCCESS	int	0
VERIFY_TIMEOUT	int	5

5.9 ConstantEmv.POIEmvCoreManager.EmvResultConstraints

Constant Name	Type	Value
APPLE_DATA	String	"appleData"
APPLE_MERCHANT	String	"appleMerchant"
APPLE_RESULT	String	"appleResult"
CVM	String	"cvm"
CVM_CONFIRMATION_CODE_VERIFIED	int	2
CVM_NO_CVM	int	0
CVM_SIGNATURE	int	1
EMV_DATA	String	"emvData"
ENCRYPT_DATA	String	"encryptData"
ENCRYPT_RESULT	String	"encryptResult"
SCRIPT_RESULT	String	"scriptResult"
SECOND_TAP_CANCEL	int	3

SECOND_TAP_FAIL	int	1
SECOND_TAP_RESULT	String	"secondTapResult"
SECOND_TAP_SUCCESS	int	0
SECOND_TAP_TIMEOUT	int	2

5.10 ConstantEmv.POIEmvCoreManager.EmvServiceConstraints

Constant Name	Type	Value
CONFIG	String	"Config"
TAG_PRMACQ_SET_DELIMITER	String	"DF02"
TAG_PRMACQ_SET_INDEX	String	"DF30"
TAG_PRMACQ_SET_KCV	String	"DF32"
TAG_PRMACQ_SET_KEY	String	"DF31"
TAG_SERVICE_SET_DATA	String	"DF19"
TAG_SERVICE_SET_DELIMITER	String	"DF01"
TAG_SERVICE_SET_ID	String	"DF16"
TAG_SERVICE_SET_MANAGEMENT	String	"DF18"
TAG_SERVICE_SET_PRIORITY	String	"DF17"
TAG_SERVICE_SET_PRMACQ	String	"DF21"
TAG_SERVICE_SET_PRMISS	String	"DF20"

5.11 ConstantEmv.POIEmvCoreManager.EmvTerminalConstraints

Constant Name	Type	Value
BYPASS_PIN_ENTRY	String	"BypassPINEntry"
CARD_HOLDER_CONFIRM	String	"CardHolderConfirm"
CONFIG	String	"Config"
DEFAULT_DDOL	String	"DefaultDDOL"
DEFAULT_TDOL	String	"DefaultTDOL"
EXCEPTION_FILE	String	"ExceptionFile"
FLOOR_LIMIT_CHECKING	String	"FloorLimitChecking"
FORCED_ACCEPT	String	"ForcedAccept"
FORCED_ONLINE	String	"ForcedOnline"
GET_DATA_FOR_PIN_COUNTER	String	"GetDataForPINCounter"
IFD_SERIAL_NUMBER	String	"IfdSerialNumber"
ISSUER_REFERRAL	String	"IssuerReferral"
LANGUAGE_SELECT	String	"LanguageSelect"
MERCHANT_CATEGORY_CODE	String	"MerchantCategoryCode"
MERCHANT_ID	String	"MerchantId"
MERCHANT_NAME	String	"MerchantName"
PSE	String	"Pse"
RANDOM_TRANSACTION_SELECTION	String	"RandomTransactionSelection"
REVOCATION_ISSUER_PUBLIC_KEY	String	"RevocationIssuerPublicKey"

SETTINGS_AMEX	int	24
SETTINGS_DISCOVER	int	23
SETTINGS_EFTPOS	int	30
SETTINGS_INTERAC	int	29
SETTINGS_JCB	int	25
SETTINGS_MASTERCARD	int	22
SETTINGS_MIR	int	26
SETTINGS_PURE	int	28
SETTINGS_RUPAY	int	27
SETTINGS_UNIONPAY	int	21
SETTINGS_VISA	int	20
SUBSEQUENT_BYPASS_PIN_ENTRY	String	"SubsequentBypassPINEntry"
TAG_AMEX_SET_ENTRY_POINT	String	"DF30"
TAG_AMEX_SET_KERNEL_CONFIG	String	"DF1B"
TAG_AMEX_SET_QUALIFIERS	String	"9F6E"
TAG_AMEX_SET_STATUS_ZERO_AMOUNT	String	"DF32"
TAG_CARD_DATA_INPUT_CAPABILITY	String	"DF8117"
TAG_DISCOVER_SET_ENTRY_POINT	String	"DF30"
TAG_DISCOVER_SET_QUALIFIERS	String	"9F66"
TAG_DISCOVER_SET_STATUS_ZERO_AMOUNT	String	"DF32"
TAG_EFTPOS_SET_ENTRY_POINT	String	"DF30"
TAG_EFTPOS_SET_KERNEL_CONFIG	String	"DF1B"
TAG_EFTPOS_SET_QUALIFIERS	String	"9F66"
TAG_EFTPOS_SET_STATUS	String	"DF31"
TAG_EFTPOS_SET_ZERO_AMOUNT	String	"DF32"
TAG_JCB_SET_ENTRY_POINT	String	"DF30"
TAG_JCB_SET_KERNEL_CONFIG	String	"DF1B"
TAG_JCB_SET_QUALIFIERS	String	"9F53"
TAG_JCB_SET_STATUS	String	"DF31"
TAG_JCB_SET_ZERO_AMOUNT	String	"DF32"
TAG_MASTERCARD_SET_CVM_CAPABILITIES	String	"DF8118"
TAG_MASTERCARD_SET_DEFAULT_UDOL	String	"DF811A"
TAG_MASTERCARD_SET_KERNEL_CONFIG	String	"DF811B"
TAG_MASTERCARD_SET_KERNEL_ID	String	"DF810C"
TAG_MASTERCARD_SET_MAGSTRIPE_APP_VERSION	String	"9F6D"
TAG_MASTERCARD_SET_MAGSTRIPE_CVM_CAPABILITIES	String	"DF811E"
TAG_MASTERCARD_SET_MAGSTRIPE_NO_CVM_CAPABILITIES	String	"DF812C"
TAG_MASTERCARD_SET_MOBILE_SUPPORT_INDICATOR	String	"9F7E"
TAG_MASTERCARD_SET_NO_CVM_CAPABILITIES	String	"DF8119"

TAG_MASTERCARD_SET_RRP_ACCURACY_THRESHOLD	String	"DF8136"
TAG_MASTERCARD_SET_RRP_CAPDU_EXPECTED	String	"DF8134"
TAG_MASTERCARD_SET_RRP_MAX_GRACE	String	"DF8133"
TAG_MASTERCARD_SET_RRP_MIN_GRACE	String	"DF8132"
TAG_MASTERCARD_SET_RRP_MISMATCH_THRESHOLD	String	"DF8137"
TAG_MASTERCARD_SET_RRP_RAPDU_EXPECTED	String	"DF8135"
TAG_MIR_SET_ENTRY_POINT	String	"DF30"
TAG_MIR_SET_QUALIFIERS	String	"9F66"
TAG_MIR_SET_STATUS_ZERO_AMOUNT	String	"DF32"
TAG_PURE_SET_ENTRY_POINT	String	"DF30"
TAG_PURE_SET_KERNEL_CONFIG	String	"DF1B"
TAG_PURE_SET_QUALIFIERS	String	"C7"
TAG_PURE_SET_STATUS	String	"DF31"
TAG_PURE_SET_ZERO_AMOUNT	String	"DF32"
TAG_SECURITY_CAPABILITY	String	"DF811F"
TAG_UNIONPAY_SET_ENTRY_POINT	String	"DF30"
TAG_UNIONPAY_SET_QUALIFIERS	String	"9F66"
TAG_UNIONPAY_SET_STATUS_ZERO_AMOUNT	String	"DF32"
TAG_VISA_SET_ENTRY_POINT	String	"DF30"
TAG_VISA_SET_KERNEL_CONFIG	String	"DF1B"
TAG_VISA_SET_QUALIFIERS	String	"9F66"
TAG_VISA_SET_STATUS_ZERO_AMOUNT	String	"DF32"
TERMINAL_CAPABILITY	String	"TerminalCapability"
TERMINAL_COUNTRY_CODE	String	"TerminalCountryCode"
TERMINAL_ENTRY_MODE	String	"TerminalEntryMode"
TERMINAL_EX_CAPABILITY	String	"TerminalExCapability"
TERMINAL_ID	String	"TerminalId"
TERMINAL_TYPE	String	"TerminalType"
TRANS_CURRENCY_CODE	String	"TransCurrencyCode"
TRANS_CURRENCY_EXP	String	"TransCurrencyExp"
TRANS_REFER_CURRENCY_CODE	String	"TransReferCurrencyCode"
TRANS_REFER_CURRENCY_EXP	String	"TransReferCurrencyExp"
TYPE_AMEX	int	7
TYPE_CONFIG	int	2
TYPE_DISCOVER	int	6
TYPE_INTERAC	int	10
TYPE_MASTERCARD	int	5
TYPE_MIR	int	8
TYPE_RUPAY	int	9
TYPE_TERMINAL	int	1

TYPE_UNIONPAY	int	4
TYPE_VISA	int	3
UNABLE_TO_GO_ONLINE	String	"UnableToGoOnline"
VELOCITY_CHECKING	String	"VelocityChecking"

5.12 ConstantEnv.POIEnvCoreManager.EnvTransDataConstraints

Constant Name	Type	Value
ACCOUNT_MASK_HEAD	String	"accountMaskHead"
ACCOUNT_MASK_TAIL	String	"accountMaskTail"
ACCOUNT_TYPE	String	"accountType"
AMOUNT_CONFIG	String	"amountConfig"
APPLE_VAS	String	"appleVas"
CL_SPECIAL_TYPE	String	"clSpecialType"
CT_SPECIAL_TYPE	String	"ctSpecialType"
ENCRYPT_BASE64	String	"encryptBase64"
ENCRYPT_CONTACT	String	"encryptContact"
ENCRYPT_CONTACTLESS	String	"encryptContactless"
ENCRYPT_EMV_DATA	String	"encryptEmvData"
ENCRYPT_KEY_INDEX	String	"encryptKeyIndex"
ENCRYPT_KEY_MODE	String	"encryptKeyMode"
ENCRYPT_KEY_MODE_TRANS_ARMOR	int	1
ENCRYPT_MAGSTRIPE	String	"encryptMagstripe"
ENCRYPT_MODE	String	"encryptMode"
ENCRYPT_MODE_CBC	int	2
ENCRYPT_MODE_ECB	int	1
ENCRYPT_OPEN_CONTACT	int	1
ENCRYPT_OPEN_CONTACTLESS	int	2
ENCRYPT_OPEN_MAGSTRIPE	int	4
ENCRYPT_PADDING	String	"encryptPadding"
ENCRYPT_SHA1	String	"encryptSHA1"
ENCRYPT_TYPE	String	"encryptType"
ENCRYPT_TYPE_DUKPT_DATA_REQUEST	int	3
ENCRYPT_TYPE_DUKPT_DATA_RESPONSE	int	4
ENCRYPT_TYPE_DUKPT_MAC	int	2
ENCRYPT_TYPE_DUKPT_PIN	int	5
ENCRYPT_TYPE_RSA	int	6
ENCRYPT_TYPE_TDK	int	1
ENCRYPT_TYPE_TTK	int	7
ENCRYPT_VECTOR	String	"encryptVector"
GOOGLE_SMART_TAP	String	"googleSmartTap"
OPEN_ENCRYPT	String	"openEncrypt"
RSA_TRANS_ARMOR_KEY_ID	String	"rsaTransArmorKeyId"

RSA_TRANS_ARMOR_POS_ID	String	"rsaTransArmorPosId"
SPECIAL_CONTACT	String	"specialContact"
SPECIAL_CONTACT_TIME	String	"specialContactTime"
SPECIAL_MAGSTRIPE	String	"specialMagstripe"
SPECIAL_MAGSTRIPE_TIME	String	"specialMagstripeTime"
SPECIAL_START_MODE	String	"specialStartMode"
SPECIAL_TYPE	String	"specialType"
START_A	int	0
START_B	int	1
START_C	int	2
START_D	int	3
TARNS_COUNTER	String	"tarnsCounter"
TRANS_AMOUNT	String	"transAmount"
TRANS_AMOUNT_OTHER	String	"transAmountOther"
TRANS_DATE	String	"transDate"
TRANS_FALLBACK	String	"transFallback"
TRANS_MODE	String	"transMode"
TRANS_TIME	String	"transTime"
TRANS_TIMEOUT	String	"transTimeout"
TRANS_TYPE	String	"transType"
USE_ABECS	String	"useABECS"
USE_CARD_READ_SUCCESS	String	"useCardReadSuccess"
USE_CT_RUPAY	String	"useCTRupay"
USE_DELAY_PIN	String	"useDelayPIN"
USE_ENCRYPT_AMEX_TRACK	String	"useEncryptAmexTrack"
USE_FILTER	String	"useFilter"
USE_FORCED_AID_SELECTION	String	"useForcedAIDSelection"
USE_FORCED_ICC_AID_SELECTION	String	"useForcedIccAIDSelection"
USE_FORCED_RETURN_OF_CARD	String	"useForcedReturnOfCard"
USE_GAC1_FILTER	String	"useGac1Filter"
USE_GAC2_FILTER	String	"useGac2Filter"
USE_GPO_BEFORE_FILTER	String	"useGpoBeforeFilter"
USE_LOG	String	"log"
USE_MAGSTRIPE_FILTER	String	"useMagstripeFilter"
USE_PPSE_FAIL_SEND_AIDS_OPTION	String	"usePPSEFailSendAidsOption"
USE_SELECT_AFTER_FILTER	String	"useSelectAfterFilter"
USE_SELECT_KERNEL	String	"useSelectKernel"
USE_SPECIAL_AID_SELECTION	String	"useSpecialAIDSelection"
USE_USA_VISA	String	"useUSAVisa"

5.13 ConstantPrinter

Constant Name	Type	Value
STATUS_IDLE	int	0

STATUS_NO_PAPER	int	3
STATUS_OVERHEAT	int	2
STATUS_PRINTING	int	1

5.14 ConstantPrinter.Align

Enum Constant	Type	Description
CENTER	ENUM	Center alignment
LEFT	ENUM	Left alignment
RIGHT	ENUM	Right alignment

5.15 ConstantPrinter.BarcodeFormat

Enum Constant	TYPE	Description
CODABAR	ENUM	CODABAR 1D format.
CODE_128	ENUM	Code 128 1D format.
CODE_39	ENUM	Code 39 1D format.
CODE_93	ENUM	Code 93 1D format.
DATA_MATRIX	ENUM	Data Matrix 2D barcode format.
EAN_8	ENUM	EAN-8 1D format.
QR_CODE	ENUM	QR Code 2D barcode format.
UPC_E	ENUM	UPC-E 1D format.

5.16 ConstantPrinter.PrintFailurePolicy

Enum Constant	TYPE	Description
ABORT_ALL	ENUM	Aborts all pending print jobs in the queue
IGNORE_AND_CONTINUE	ENUM	Ignores current error and continues next print

5.17 ConstantPrinter.FontSize

Constant Name	Type	Value
LARGE	float	36.0f
NORMAL	float	24.0f
SMALL	float	16.0f

5.18 ConstantSecurity

Constant Name	Type	Value
AUTHENTICATION_BOTH	int	2
AUTHENTICATION_GENERATION	int	0
AUTHENTICATION_VERIFICATION	int	1
DUKPT_KEY_SELECT_DATA_REQUEST	int	1
DUKPT_KEY_SELECT_DATA_RESPONSE	int	2

DUKPT_KEY_SELECT_MAC_REQUEST_OR_RESPONSE	int	0
DUKPT_KEY_SELECT_PIN_ENCRYPTION	int	3
DUKPT_MAC_MODE_CBC	int	2
DUKPT_MAC_MODE_ECB	int	0
DUKPT_MODE_AES_MODE	int	128
ENCRYPTION_ALGORITHM_AES	int	16
ENCRYPTION_ALGORITHM_TDES	int	0
ENCRYPTION_MECHANISM_DUKPT	int	2
ENCRYPTION_MECHANISM_MK_SK	int	1
KEY_ALG_TYPE_2TDEA	int	0
KEY_ALG_TYPE_3TDEA	int	16
KEY_ALG_TYPE_AES_128	int	32
KEY_ALG_TYPE_AES_192	int	48
KEY_ALG_TYPE_AES_256	int	64
KSN_AUTO_INCREASING_BY_DUKPT_TDES_MAC_BOTH_KEY	int	0
KSN_NOT_AUTO_INCREASING_BY_DUKPT_TDES_MAC_BOTH_KEY	int	20
KSN_NOT_AUTO_INCREASING_BY_DUKPT_TDES_MAC_RSP_KEY	int	40
MAC_ALGORITHM_ANSI_X9_19	int	2
MAC_ALGORITHM_ANSI_X9_9	int	3
MAC_ALGORITHM_CBC	int	0
MAC_ALGORITHM_XOR_ECB_MAC	int	1
MAC_MODE_ANSI_X9_19	int	2
MAC_MODE_ANSI_X9_9	int	3
MAC_MODE_CBC	int	0
MAC_MODE_XOR_ECB_MAC	int	1
NOT_SELF_INCREASING	int	0
OPERATION_DIRECTION_DECRYPT	int	0
OPERATION_DIRECTION_ENCRYPT	int	1
OPERATION_MODE_CBC	int	2
OPERATION_MODE_ECB	int	0
PED_CALC_DES_MODE_CBC_DEC	int	2
PED_CALC_DES_MODE_CBC_ENC	int	3
PED_CALC_DES_MODE_ECB_DEC	int	0
PED_CALC_DES_MODE_ECB_ENC	int	1
PED_CALC_DUKPT_MODE_DEC	int	0
PED_CALC_DUKPT_MODE_ENC	int	1
PED_CALC_RSA_MODE_NO_PADDING	int	0
PED_CALC_RSA_MODE_OAEP_PADDING	int	2
PED_CALC_RSA_MODE_PKCS1_PADDING	int	1
PED_TAK	int	4
PED_TDK	int	5
PED_TEK	int	6
PED_TIK	int	7
PED_TLK	int	1

PED_TMK	int	2
PED_TPK	int	3
PED_TTK	int	9
PINBLOCK_DUKPT_FMT_ISO9564_0	int	32
PINBLOCK_DUKPT_FMT_ISO9564_0_KSN_INC	int	0
PINBLOCK_DUKPT_FMT_ISO9564_1	int	33
PINBLOCK_DUKPT_FMT_ISO9564_1_KSN_INC	int	1
PINBLOCK_DUKPT_FMT_ISO9564_2	int	34
PINBLOCK_DUKPT_FMT_ISO9564_2_KSN_INC	int	2
PINBLOCK_DUKPT_FMT_ISO9564_4	int	36
PINBLOCK_DUKPT_FMT_ISO9564_4_KSN_INC	int	4
PINBLOCK_TPK_FMT_ISO9564_0	int	0
PINBLOCK_TPK_FMT_ISO9564_1	int	1
PINBLOCK_TPK_FMT_ISO9564_3	int	2
PINBLOCK_TPK_FMT_ISO9564_4	int	4
SELF_INCREASING	int	64

5.19 ConstantScanner

Constant Name	Type
ALL_BARCODES	ConstantScanner.BarcodeFormat[]
ONE_DIMENSIONAL_BARCODES	ConstantScanner.BarcodeFormat[]
TWO_DIMENSIONAL_BARCODES	ConstantScanner.BarcodeFormat[]

5.20 ConstantScanner.BarcodeFormat

Enum Constant	Type	Description
CODABAR	1D	Used in blood banks and libraries, supports digits and some special characters
CODABLOCKF	1D	Stacked barcode format used in logistics and industrial applications
CODE11	1D	Primarily for telecom equipment, supports digits and hyphens
CODE128	1D	High-density linear barcode supporting ASCII, widely used in logistics and retail
CODE39	1D	General-purpose alphanumeric barcode for industrial applications
CODE93	1D	Improved version of Code 39 with higher density character support
EAN13	1D	International standard retail barcode (13-digit)
EAN8	1D	Compact version of EAN-13 for small products (8-digit)
GS1_128	1D	Supply chain barcode based on Code 128 standard
GS1_DATABAR	1D	Compact barcode for small retail items like produce
HK25	1D	Hong Kong variant of Interleaved 2 of 5 barcode
IATA25	1D	Air cargo specific barcode format
INDUSTRIAL25	1D	Industrial variant of 25 barcodes
ITF25	1D	High-density numeric barcode for carton labeling
MATRIX25	1D	Variant of 25 barcode format
MSI	1D	Inventory management barcode (digits only)
TELEPEN	1D	UK barcode standard supporting full ASCII
UPCA	1D	North American retail product barcode (12-digit)

UPCE	1D	Compressed version of UPC-A for small packages
USPS4ST	1D	US Postal Service tracking barcode
AZTEC	2D	Compact matrix barcode suitable for small spaces
DATAMATRIX	2D	2D barcode for product identification and marking
DOTCODE	2D	Dot-based barcode for high-speed industrial printing
GRIDMATRIX	2D	Chinese-developed 2D barcode for Chinese characters
GS1_DATAMATRIX	2D	GS1-compliant version of Data Matrix
HANXIN	2D	Chinese national standard 2D barcode
MAXICODE	2D	Fixed-size matrix barcode used in logistics
MICROPDF	2D	Compact PDF417 variant for ID documents
PDF417	2D	Stacked 2D barcode for transportation and ID cards
QRCODE	2D	Popular matrix barcode for general-purpose use

5.21 ConstantScanner.ScannerCameraType

Enum Constant	TYPE	Description
CAMERA_FRONT	ENUM	Front Camera
CAMERA_REAR	ENUM	Rear Camera
SCANNER	ENUM	E-Series Scanner

5.22 ConstantScanner.BarcodeSupport

Enum Constant	TYPE	Description
ALL_SUPPORT	ENUM	This barcode type is supported by both cameras and scanner
CAMERA_SUPPORT	ENUM	This barcode type is only supported by front/rear cameras
SCANNER_SUPPORT	ENUM	This barcode type is only supported by scanner

● 6. Access permission

Permission Name	Support Status
com.kozen.perm.MAGNETIC	Not yet support
com.kozen.perm.CONTACT	Not yet support
com.kozen.perm.CONTACTLESS	Not yet support
com.kozen.perm.PINPAD	Not yet support
com.kozen.perm.SECURITY	Not yet support
com.kozen.perm.LED	Not yet support
com.kozen.perm.PRINTER	Not yet support
com.kozen.perm.SCANNER	Not yet support